# Selling Down-Market: Market Access and the Quality of Chinese Exports

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**Abstract**

Trade liberalization can change export quality through cheaper inputs or through access to foreign buyers. I study the latter, exploiting the 2001 US grant of permanent normal trade relations to China, which eliminated the threat of tariff increases that varied across products. Comparing the same firm's US and non-US sales of a product before and after 2002, I find that quality fell about 4.5 percent more in highly exposed products than in barely exposed ones. The fall appears on impact, does not detectably reverse, and has no counterpart in the EU. Prices fell, but quantities rose less than the price decline alone predicts. I develop a model in which access lets firms sell down-market, to buyers who value quality less.

**JEL Codes:** F12, F13, F14, L15
**Keywords:** export quality, market access, trade policy uncertainty, permanent normal trade relations

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## 1. Introduction

Improved market access is generally expected to raise export quality. Richer buyers reward better goods (Fajgelbaum, Grossman, and Helpman, 2011; Verhoogen, 2008), better goods sort toward the destinations that pay for them (Crozet, Head, and Mayer, 2012), and larger markets raise the return to the costly investments that quality requires (Kugler and Verhoogen, 2012). Yet a market that opens can also bring firms to customers they did not previously serve, and those customers may care less about quality than the customers already reached. This paper asks whether improved access raises or lowers the quality of the products firms already sell.

I study China's accession to the World Trade Organization in December 2001 and the permanent normal trade relations (PNTR) treatment that applied to its exports to the United States from January 2002.[1] Before then, the United States renewed China's normal trade relations status annually. A failed renewal would have returned Chinese goods to the much higher column-2 tariff schedule. PNTR removed this threat but did not change the tariff China actually paid. The gain in policy certainty differed across products because the gap between the normal and column-2 schedules differed across products. I use that NTR gap as exposure to the change in market access.

I measure quality as the demand a seller commands at a given price, relative to other sellers of the same product (Khandelwal, 2010; Khandelwal, Schott, and Wei, 2013). After PNTR, the quality of continuing firms' US sales fell about 4.5 percent more in highly exposed products than in barely exposed ones, comparing the 90th with the 10th percentile of the NTR gap and relative to the same firms' sales of those products elsewhere.[2]

I use Chinese customs records for 2000–2014 and focus on continuing export relationships. The empirical design is a triple difference: it asks whether the US sales of a firm and HS6 product changed after 2002, relative to that firm's sales of the same product outside the United States, by more in products with larger NTR gaps. The baseline specification compares a firm's US and non-US sales of a product within a year, using the roughly 12 percent of continuing firm–products that sell to both the United States and elsewhere.[3] This comparison holds fixed shocks common to a firm's product across destinations, such as changes in input costs, technology and China-side policies, and focuses on the market-access margin rather than the production-cost margin of trade liberalization.

Three further facts characterize the result. First, the decline begins in 2002, when PNTR takes effect, and does not detectably reverse. Second, prices fall and quantities rise. In the baseline specification, prices fall by 5.0 percent across the exposure range and quantities rise by 7.7 percent. Third, the result is specific to the United States: the corresponding quality estimates for the EU-15, where China already had permanent most-favored-nation status, are close to zero, and the US estimate survives controls for changes in US quotas on textiles and apparel.[4]

These price and quantity responses are central to the interpretation. The quality index infers higher quality when a seller receives a higher price or sells more quantity than comparable sellers of the same product. For an

[1] PNTR was signed into law in October 2000, conditional on China's accession. Since 2001 may be partly treated, I also report estimates with 2000 alone as the base year.

[2] The 10th-to-90th percentile difference in the NTR gap is 0.387.

[3] The broad specification replaces the firm × product × year effects with firm × year and product × year effects, so it uses every continuing relationship (Tables 1 and 2).

[4] The European Union granted China permanent most-favored-nation status in 1985 (Feng, Li, and Swenson, 2017). With US rows dropped, the EU-15 coefficient is +0.010 (0.032).

unchanged product, a lower price accompanied by the implied movement along the demand curve would leave this index unchanged. With an elasticity of substitution of five, the 5.0 percent price decline would predict a quantity increase of about 25 log points, or roughly 29 percent. Quantity rose 7.7 percent, so the combination of lower prices and limited quantity growth implies lower quality. Under the CES demand that underlies the index, a pure reduction in marginal cost or markup on an unchanged product would bring a larger quantity response than the one observed (Section 5).

To interpret these facts, I develop a model of selling down-market. Better access lets firms sell further down the US market. If the new buyers there value quality less, firms can lower quality to serve them. In the model a seller chooses both the quality of a destination-specific product and the range of buyers it serves; quality is costly to provide, and buyers differ in how much they value it. The model also shows why the quality index need not record the full fall in quality: the additional sales from broader access offset part of it. The model can produce the facts, but the customs data do not reveal which buyers firms reach.

More secure access to an export market lowered the quality of products that Chinese firms were already selling. The decline is similar in magnitude, but opposite in sign, to the quality increase that lower input tariffs bring in Fan, Li, and Yeaple (2015).[5] My baseline estimate holds fixed shocks common to a firm's product across its US and non-US sales, including changes in input costs. Lower input tariffs reduce the cost of supplying quality, whereas improved access changes which buyers firms can profitably serve.

Most findings that liberalization raises export quality concern a country's own liberalization: cheaper imported inputs raise it (Fan, Li, and Yeaple, 2015; Bas and Strauss-Kahn, 2015; Fieler, Eslava, and Xu, 2018), and tougher competition at home raises it for varieties near the world frontier (Amiti and Khandelwal, 2013). On the access side, upgrading comes from selling to demanding buyers, as with Mexican plants exporting after the peso devaluation (Verhoogen, 2008), Egyptian rug producers given access to high-income markets (Atkin, Khandelwal, and Osman, 2017), and Portuguese firms buying better inputs to sell to richer destinations (Bastos, Silva, and Verhoogen, 2018). PNTR differs because it opened a whole market, including buyers who care less about quality.

The trade-policy-uncertainty literature has studied export entry, trade volumes and employment. Pierce and Schott (2016) measure the threat product by product with the NTR gap. Handley (2014) explains why a threat that never materialized still mattered: entering a market costs money a firm cannot recover, so a firm holds back while a tariff increase could make the market unprofitable. Handley and Limão (2017) apply this reasoning to China and show that uncertainty about future tariffs discourages export investment. Existing evidence on quality as China's access improved concerns the turnover of sellers. Khandelwal, Schott, and Wei (2013) attribute the fall in prices after quota removal to net entry, and Feng, Li, and Swenson (2017) find that entrants under PNTR sold higher quality at lower prices than the firms that exited. Turnover moves a market's average quality with no seller having changed anything. I instead study what happened to products that a seller was already shipping.

The rest of the paper proceeds as follows. Section 2 describes PNTR and the customs data. Section 3 presents the empirical design. Section 4 reports the main results. Section 5 considers alternative explanations and robustness checks. Section 6 develops the model. Section 7 concludes.

[5] At $\sigma = 5$ and on a closely related measure, their average input-tariff cut raised quality by 0.046 in logs, about 4.7 percent (Section 4.5).

## 2. Setting and data

### 2.1. PNTR and product-level exposure

Before 2002 the United States renewed China's normal trade relations status once a year. A failed renewal would have returned Chinese goods to the much higher column-2 tariff schedule, whose rates were largely set under the Smoot–Hawley Tariff Act of 1930. PNTR was signed into law in October 2000, conditional on China's accession to the WTO. Accession came in December 2001, and PNTR applied from 1 January 2002 (Pierce and Schott, 2016). I treat 2002 as the first post-PNTR year.

PNTR did not change the NTR tariff China paid; it removed the threat of reversion to column-2 rates, an uncertainty that Handley and Limão (2017) show discouraged entry investment. The reduction in uncertainty was larger for products with a wider gap between the two schedules. Following Pierce and Schott (2016), I measure exposure by this NTR gap, the column-2 rate minus the 2000 NTR rate.

Figure 1 plots the NTR gap across the 3,794 products in the estimation sample. Its median is 0.27, and its 10th and 90th percentiles are 0.06 and 0.45, implying a 10th-to-90th percentile difference of about 0.39. I report economic magnitudes for an increase in exposure of this size.

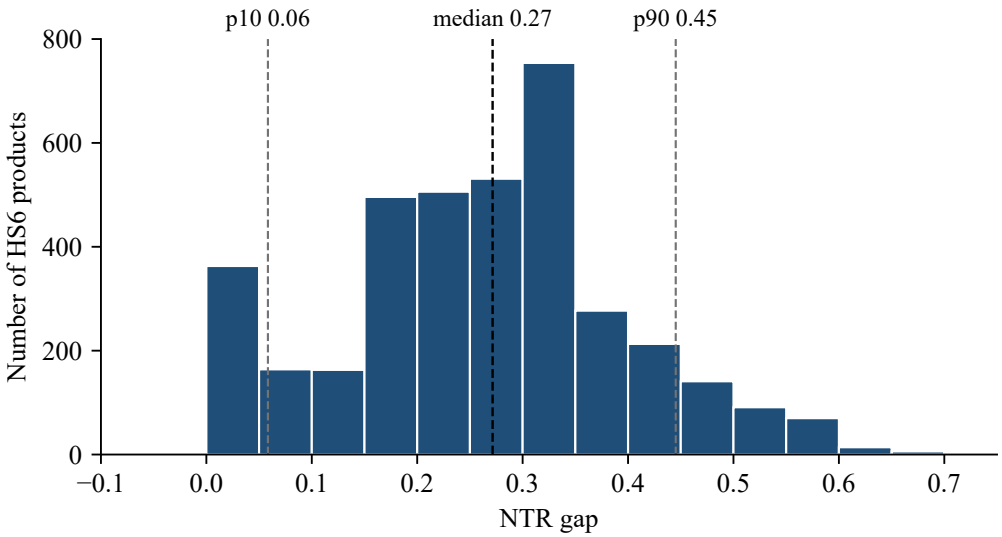



**Figure 1:** Exposure to PNTR across products

*Notes:* Distribution of the NTR gap, as a proportion, across the 3,794 HS6 products of the estimation sample, in bins of 0.05. Dashed lines mark the 10th percentile 0.06, the median 0.27 (black) and the 90th percentile 0.45 (maximum 0.67). The five products with a slightly negative gap are in the two bins below zero.

### 2.2. Chinese customs data

The data are the export records of the Chinese customs administration for 2000 to 2014, which I obtain from the RESSET database. Each record is a firm, an eight-digit (HS8) tariff line, a destination, a trade regime and a year, with the value shipped and the quantity. Ordinary and processing trade are pooled. I aggregate the records to a firm, an HS6 product, a destination group and a year. The destination group is either the United States or a pooled non-US group containing every other destination the firm serves with that product in that year. Within a

firm, product, destination group and year, I sum values and quantities across records and define price as total value divided by total quantity (the unit value). Because aggregation requires a common physical unit, I retain, within each product and year, the unit accounting for the greatest export value.

Reported quantities for 2006 are anomalous, so I treat 2006 as missing for price, quantity and quality. Within the same firm, HS8 line and destination, unit values jump from 2005 to 2006 and reverse in 2007 while export values move little. BACI unit values for China-to-US exports show no such jump, and the discrepancy differs between US and non-US shipments, so product-by-year effects cannot absorb it (Appendix D.4). The 2002–04 window does not include 2006.

### 2.3. Continuing export relationships

I restrict the analysis to continuing firm–product–destination-group relationships. A relationship is a firm's shipments of one HS6 product to one destination group. It is continuing if it is observed both before and after 2002. The estimates therefore describe adjustment within relationships that survive the policy change; they do not estimate PNTR's overall effect on the composition of exporters. I examine entry and exit separately in Section 5.

Table 1 describes the two samples that carry the results. The first column is every continuing relationship, the sample of the broad specification of Section 3. It covers 592,473 firm–products and 3,794 products. The second column holds the firm–products that identify the baseline specification of Section 3. There are 69,030 of them, about 12 percent of firm–products, and they cover 2,503 of the 3,794 products.

**Table 1:** Estimation samples

| | Broad<br>All continuing firm–products | Baseline<br>Firm–products that identify it, 2002–04 |
|---|---|---|
| *Panel A. Coverage* | | |
| Firm–product–destination–year rows | 3,485,986 | 779,854 |
| Firm–products | 592,473 | 69,030 (12%) |
| HS6 products | 3,794 | 2,503 (66%) |
| Rows in the column's own regression | 3,479,900 | 836,554 |
| *Panel B. Pre-period means* | | |
| Unit value (log) | 1.01 | 0.98 |
| Quantity (log) | 9.06 | 10.19 |
| US share of the firm's exports (per firm) | 0.17 | 0.33 |
| NTR gap (per product) | 0.26 | 0.28 |

*Notes:* Continuing export relationships, 2006 excluded. The broad column is every continuing relationship; the baseline column holds the firm–products that identify the 2002–04 estimate (Section 3). In the baseline column, rows count only the years in which a firm–product sells to both groups. The broad regression uses 6,086 fewer rows than the first row reports; the baseline regression also uses the firm–products that sell to both groups in some year but do not identify the 2002–04 estimate. In Panel B, unit value and quantity are averaged across 2000–01 observations, the US share across firms and the NTR gap across products.

### 2.4. Measuring quality

A product that sells more than its price alone would predict, relative to other sellers of the same product, is inferred to be of higher quality. This approach follows Khandelwal (2010) and Khandelwal, Schott, and Wei (2013). Under CES demand with elasticity of substitution $\sigma$ across sellers of a product, inferred quality is, up to a product-by-destination-by-year term common to all sellers,

$$\ln \lambda_{fhdt} = \frac{\sigma}{\sigma - 1} \ln p_{fhdt} + \frac{1}{\sigma - 1} \ln q_{fhdt}, \tag{1}$$

where $f$ is the firm, $h$ the HS6 product, $d$ the destination group, $t$ the year, $p$ the price and $q$ the quantity. Quality rises with price, because a seller that holds its quantity while charging more must offer something buyers value more. It rises with quantity at a given price for the same reason. Quality in this sense is the value buyers place on a product and need not map one-for-one into any particular physical attribute.

I set $\sigma = 5$, a value within the range of estimates of substitution across sellers within product markets. At this value, $\ln \lambda = 1.25 \ln p + 0.25 \ln q$.[6] Section 4 reports results for alternative values of $\sigma$. When $\sigma$ is common across products, the coefficient on quality is mechanically $\sigma/(\sigma - 1)$ times the price coefficient plus $1/(\sigma - 1)$ times the quantity coefficient.

## 3. Empirical design

Identification comes from a triple difference: between US and non-US destinations, before and after 2002, and across products with different NTR gaps. The coefficient of interest measures whether the post-2002 change in a firm's US-relative outcome is larger for products more exposed to PNTR.

### 3.1. The estimating equation

For each specification, I estimate

$$y_{fhdt} = \beta_1 \, \text{Gap}_h \times \text{US}_d \times \mathbf{1}[2002 \le t \le 2004] + \beta_2 \, \text{Gap}_h \times \text{US}_d \times \mathbf{1}[t \ge 2005] + \text{fixed effects} + \varepsilon_{fhdt}, \tag{2}$$

where $y$ is quality, log price or log quantity for firm $f$, HS6 product $h$, destination group $d$ (the United States or the pooled non-US group) and year $t$. $\text{Gap}_h$ is the product's NTR gap and $\text{US}_d$ marks the United States. The fixed effects differ across specifications and are listed in Table 2. The two post-PNTR interactions enter the same regression, and their coefficients are relative to the 2000–01 pre-PNTR period. A separate regression replaces the two period terms with one for all years from 2002 to 2014, and gives the pooled estimate. Because PNTR was signed into law in October 2000, expectations may have moved before the formal change, so 2001 may be partly treated. I therefore also report estimates that use 2000 alone as the pre-PNTR base year (Section 5). Year 2006 is

[6] Firm-level estimates include a median of 3.9 across firms within product groups in Hottman, Redding, and Weinstein (2016), 4.26 in Piveteau and Smagghue (2019), and about 5 in Fontagné, Martin, and Orefice (2018). Felbermayr and Sandkamp (2020) estimate an elasticity of −4.48 of Chinese firms' US shipments with respect to anti-dumping duties, with an insignificant price response; under constant elasticity of substitution, that response implies a $\sigma$ of about 4.5. Khandelwal, Schott, and Wei (2013) use 4, and Fan, Li, and Yeaple (2015) use 5 and 10. Soderbery (2018) estimates US import demand elasticities by HS4 product, treating each source country's exports of a product as one variety. These measure substitution between countries of origin rather than between firms, and they are lower: mapped to the products here, their median is 2.9. Section 4.4 reports estimates that use them product by product.

omitted, as Section 2 explains. I cluster standard errors by HS6 product, the level at which exposure varies, and report two-way clustering by firm and HS6 product in the appendix.

I focus on 2002–04 because the estimated responses of relationship survival and of selling in both destination groups are small in 2002–04 and grow after 2005 (Section 5). The 2005–14 interaction captures the longer-run evolution of the effect, including any reversal.

### 3.2. Two specifications

The specifications differ in their fixed effects, which Table 2 lists. The broad specification includes four sets: firm × year, product × year, destination group × year and firm × product × destination group. It compares each relationship with its own average before 2002, relative to the same firm's other relationships in that year, and uses all continuing relationships.

The baseline specification replaces the firm × year and product × year effects with firm × product × year effects, which nest them. It compares a firm's own US and non-US shipments of a product in the same year. The cost is coverage: the 2002–04 estimate is identified from firm–products that ship to both the United States and the non-US group in the same year in at least one pre-PNTR year and at least one year between 2002 and 2004, though not necessarily in every year. About 12 percent of firm–products meet that condition (Table 1). The baseline estimates therefore apply to continuing firm–products that serve both destination groups around the policy change. Section 5 examines whether exposure changes which relationships continue and which firm–products sell to both groups.

**Table 2:** Fixed effects by specification

| | Absorbed variation | Broad | Baseline |
|---|---|---|---|
| *Fixed effects* | | | |
| Firm × year | firm-year components common across the firm's products and destination groups | Yes | — |
| HS6 × year | product-year components common across firms and destination groups | Yes | — |
| Destination group × year | destination-group-year components common across firms and products | Yes | Yes |
| Firm × HS6 × destination group | time-invariant differences specific to each firm-product-destination-group relationship | Yes | Yes |
| Firm × HS6 × year | firm-product-year components common to the US and non-US destination groups | — | Yes |

*Notes:* Each column reports the fixed effects in one specification of eq. (2). The second column describes the additive variation each fixed effect absorbs. The broad specification compares each firm–product–destination-group relationship with its own pre-PNTR average. The baseline specification compares a firm's US and non-US shipments of the same product in the same year. Neither specification includes $\text{Gap}_h \times \text{US}_d$ separately, because the relationship fixed effect absorbs this time-invariant interaction.

In the baseline specification, firm × product × year fixed effects leave only within-year differences between a firm's US and non-US sales of the same HS6 product. Identification comes from whether that difference changes more after PNTR for products with larger NTR gaps. These effects absorb shocks common to a firm's product across destination groups within a year, including changes in input costs, the firm's technology and China's own

policies toward the product. The broad specification uses all continuing relationships, but because it does not compare destination groups within a year, such firm–product shocks remain in its error term.

### 3.3. Identification

Interpreting the coefficients causally requires that, absent PNTR, the US-minus-non-US outcome for more- and less-exposed products would have evolved similarly. A violation needs a force that changes a firm's US shipments of a product relative to its other shipments of that product, in proportion to the NTR gap, and that starts in 2002. In the baseline specification the remaining concern is therefore not a China-side cost shock common to a firm's destinations, but a US-specific product shock that is correlated with the gap and coincides with PNTR. In the broad specification, product × year effects absorb product-year shocks common across firms and destination groups. Destination group × year effects, included in both specifications, absorb shocks common across firms and products within the United States or the pooled non-US group. Because the non-US group is pooled, its composition can change over time; Section 5 re-estimates the design with each non-US destination block as a separate observation.

Because the customs data begin in 2000, the only pre-PNTR comparison is 2001 against 2000. I cannot reject that the two years are the same (Section 4.2), but one year is a limited test, so the causal interpretation rests on several further pieces of evidence: the timing of the break in 2002, the absence of a comparable response in the EU-15 (Section 5), and the comparison of a firm's US and non-US sales of the same product.

## 4. Results

### 4.1. The main estimates

In 2002–04, quality in US sales fell relative to non-US sales by 0.143 per unit of exposure in the broad specification (standard error 0.035), and by 0.118 in the baseline specification (0.038). Table 3 reports both. Across the 10th-to-90th percentile range of exposure, the US–non-US quality difference falls by 5.4 percent in the broad specification and by 4.5 percent in the baseline specification.[7]

The later period, 2005–14, gives larger coefficients (−0.298 in the broad specification, −0.230 in the baseline specification), and the pooled estimates for 2002–14 lie between the two periods.[8] Within two-digit sectors, however, the 2005–14 estimates are smaller and not significant, and they cannot be told apart from the 2002–04 estimates (Table A6, Panel B). The evidence therefore shows that quality fell and has not detectably come back, not that the fall deepened.

From here on the text reports the baseline estimates, because they also control for shocks common to a firm's product across destinations. The broad estimates are reported alongside them in the tables and figures. The two are discussed together only for prices and quantities and for the elasticity of substitution, where they differ in ways that matter for the interpretation.

[7] A coefficient $b$ converts to a percentage over the 10th-to-90th percentile gap of 0.387 as $100\,(1 - e^{b \times 0.387})$.

[8] With 2006 retained, the 2005–14 baseline estimate is −0.094 (0.080) (Appendix D.4).

**Table 3:** PNTR exposure and US sales relative to other destinations

| | (1) Broad | (2) Baseline |
|---|---|---|
| *Panel A. Quality* | | |
| 2002–04 | −0.143*** | −0.118*** |
| | (0.035) | (0.038) |
| 2005–14 | −0.298*** | −0.230*** |
| | (0.075) | (0.073) |
| 2002–14, pooled | −0.200*** | −0.160*** |
| | (0.042) | (0.043) |
| *Panel B. Unit value* | | |
| 2002–04 | −0.117*** | −0.132*** |
| | (0.030) | (0.031) |
| 2005–14 | −0.370*** | −0.368*** |
| | (0.073) | (0.075) |
| 2002–14, pooled | −0.210*** | −0.222*** |
| | (0.039) | (0.039) |
| *Panel C. Quantity* | | |
| 2002–04 | +0.013 | +0.191** |
| | (0.077) | (0.093) |
| 2005–14 | +0.660*** | +0.921*** |
| | (0.166) | (0.205) |
| 2002–14, pooled | +0.250*** | +0.467*** |
| | (0.090) | (0.119) |
| Observations | 3,479,900 | 836,554 |

*Notes:* Each entry is the coefficient on NTR gap × US × period in eq. (2), with the standard error clustered by HS6 product in parentheses. Panel A is quality at $\sigma = 5$, Panel B the log unit value and Panel C the log quantity. The 2002–04 and 2005–14 rows come from one regression; the pooled row is a separate regression over 2002–14. The sample is continuing export relationships, with 2006 excluded; the base period is the average of 2000 and 2001. Column (1) is the broad specification and column (2) the baseline one, both defined in Section 3; Table 2 lists their fixed effects. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

### 4.2. Timing

The decline appears in 2002 and does not detectably reverse. Figure 2 plots the coefficient for each year against the average of 2000 and 2001. The coefficients turn negative in 2002; before that, 2001 is not distinguishable from 2000 (the difference is +0.031, $p = 0.60$). The point estimates remain negative in every year from 2002 to 2014. After 2007 the year-by-year standard errors are 0.11 to 0.17, so the estimates for individual later years are imprecise. Figure A1 repeats the event study within two-digit sectors, and Figure A2 shows unit value and quantity by year.[9]

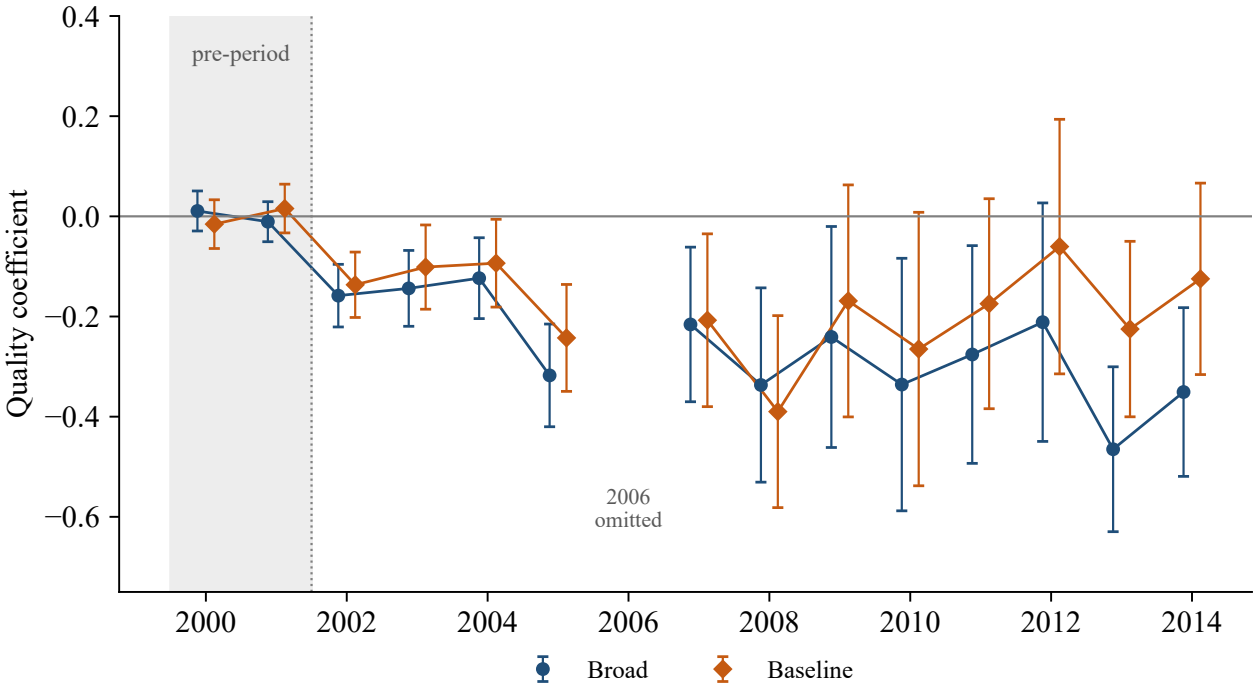



**Figure 2:** Quality of US sales relative to other destinations, by year

*Notes:* Coefficients on NTR gap × US × year from the broad and baseline specifications, relative to the average of 2000 and 2001, with 90 percent intervals. Quality at $\sigma = 5$. The two pre-period points are each half of the 2001-minus-2000 difference ($p = 0.66$ in the broad specification, 0.60 in the baseline specification). Continuing relationships; 2006 is omitted.

### 4.3. Prices and quantities

Prices fell, and quantities rose by less than they would have for an unchanged product. In the baseline specification the 2002–04 price coefficient is −0.132 (standard error 0.031), a fall of 5.0 percent across the exposure range, and the quantity coefficient is +0.191 (0.093), a rise of 7.7 percent (Table 3, Panels B and C). If the product were unchanged, a price fall of that size would move buyers along a demand curve with elasticity $\sigma = 5$, and quantity would rise by $\sigma$ times the log price fall, about 29 percent. Quantity rose by 7.7 percent. The shortfall of about 21 percentage points is what the quality measure reads as a fall in quality: what firms sold at the lower price was

[9] Within two-digit sectors the intervals from 2007 on are wide and include zero, and the point estimates are positive in 2011, 2012 and 2014. In the baseline specification within two-digit sectors, the 2001 point lies below 2000 (−0.072, $p = 0.38$), and the difference is in quantity, not in price (quantity −0.288, $p = 0.05$; price 0.000).

something buyers valued less. In the broad specification the price falls by about 4 percent (−0.117) and quantity does not rise detectably (+0.013, standard error 0.077).

Figure 3 shows the same reading for any common $\sigma$. The dashed line marks the combinations of price and quantity responses along which quality is unchanged at $\sigma = 5$. The 2002–04 points lie below that line, on the side of lower quality. A common $\sigma$ turns the pair of coefficients into a quality coefficient, and quality falls for every $\sigma$ above −(quantity coefficient)/(price coefficient). Section 4.4 reports the resulting thresholds for each specification.

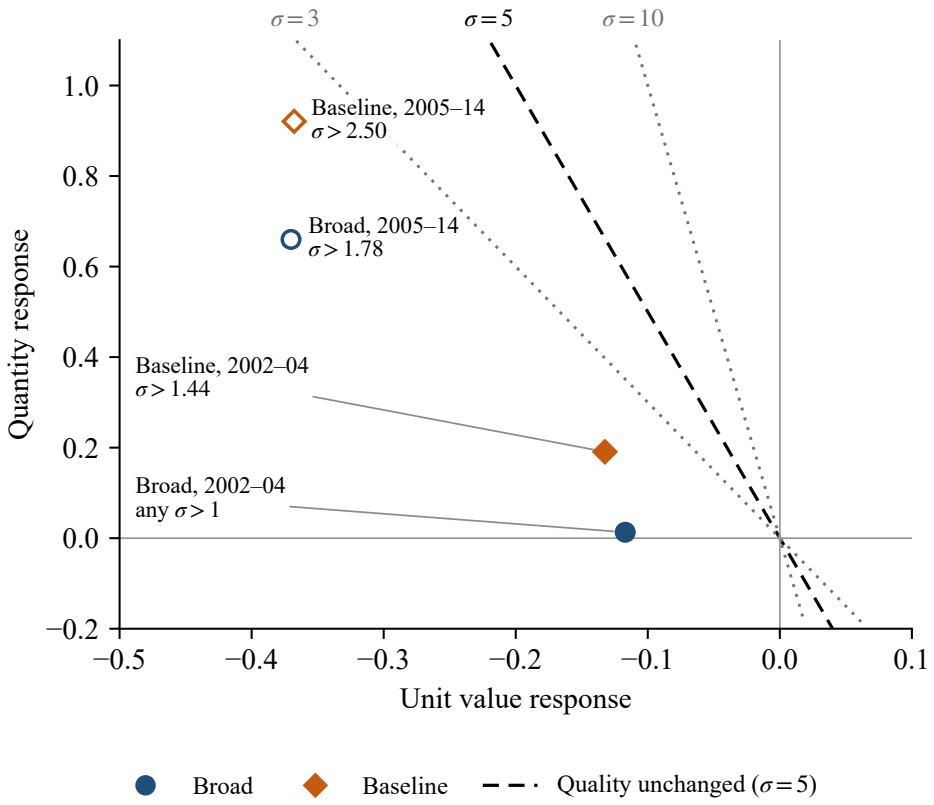



**Figure 3:** Prices fall without the matching rise in quantity

*Notes:* Each point is a pair of coefficients, the response of log price (unit value) and of log quantity, in the broad and baseline specifications. Along the dashed line quality is unchanged at $\sigma = 5$; the dotted lines show $\sigma = 3$ and $\sigma = 10$. The vertical distance from the dashed line to a point is four times its quality coefficient. A label gives the common $\sigma$ above which the point implies a fall in quality; it reads "any $\sigma > 1$" where that threshold is below 1, the lower bound that CES demand requires. Coefficients are from Table 3; filled points are 2002–04 and hollow points 2005–14.

### 4.4. The elasticity of substitution

Table 4 varies the common elasticity of substitution from $\sigma = 3$ to $\sigma = 10$ and adds a column using the product-level elasticities of Soderbery (2018), whose 5th and 95th percentiles on these products are 2.0 and 5.0. The broad estimate is negative and significant at the 1 percent level in every column. The baseline estimate is significant at the 1 percent level for $\sigma \geq 4$ but weakens at $\sigma = 3$ and with the Soderbery elasticities. In the broad specification, the 2002–04 coefficient lies between −0.129 and −0.169. The baseline coefficient is −0.113 at

$\sigma = 4$ and −0.126 at $\sigma = 10$. At $\sigma = 3$ it is −0.103 (significant at 10 percent) and with the Soderbery elasticities −0.093 (standard error 0.058, $p \approx 0.11$). It weakens because the baseline specification also has a quantity rise, and a quantity rise offsets a price fall more the lower is $\sigma$. At the baseline point estimates, quality falls for any common $\sigma$ above about 1.4, well below the elasticities reviewed in Section 2. For the broad specification, the threshold is below 1, the lower bound required by CES demand, so its point estimates imply a fall in quality for every admissible $\sigma$. The remaining results use $\sigma = 5$.

**Table 4:** Quality at different elasticities of substitution

| | (1) Soderbery (2018) | (2) $\sigma = 3$ | (3) $\sigma = 4$ | (4) $\sigma = 5$ | (5) $\sigma = 7$ | (6) $\sigma = 10$ |
|---|---|---|---|---|---|---|
| *2002–04* | | | | | | |
| Broad | −0.162*** | −0.169*** | −0.152*** | −0.143*** | −0.134*** | −0.129*** |
| | (0.049) | (0.047) | (0.038) | (0.035) | (0.032) | (0.031) |
| Baseline | −0.093 | −0.103* | −0.113*** | −0.118*** | −0.123*** | −0.126*** |
| | (0.058) | (0.054) | (0.043) | (0.038) | (0.034) | (0.033) |
| *2005–14* | | | | | | |
| Broad | −0.225** | −0.225** | −0.274*** | −0.298*** | −0.322*** | −0.338*** |
| | (0.099) | (0.091) | (0.079) | (0.075) | (0.073) | (0.072) |
| Baseline | −0.084 | −0.091 | −0.183** | −0.230*** | −0.276*** | −0.306*** |
| | (0.105) | (0.093) | (0.078) | (0.073) | (0.071) | (0.071) |
| *2002–14, pooled* | | | | | | |
| Broad | −0.185*** | −0.190*** | −0.196*** | −0.200*** | −0.203*** | −0.205*** |
| | (0.057) | (0.053) | (0.045) | (0.042) | (0.040) | (0.039) |
| Baseline | −0.089 | −0.099* | −0.140*** | −0.160*** | −0.181*** | −0.194*** |
| | (0.066) | (0.060) | (0.047) | (0.043) | (0.039) | (0.038) |

*Notes:* Coefficient on NTR gap × US × period, with the standard error clustered by HS6 product in parentheses; continuing relationships, 2006 dropped. Column (1) uses the HS4 import demand elasticities of Soderbery (2018) (median 2.9 on these products). Columns (2) to (6) use one $\sigma$ for every product; column (4) is the value used in every other table. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

### 4.5. The size of the decline

A fall of 4.5 percent is similar in magnitude, but opposite in sign, to the within-firm effect of input-tariff cuts in Fan, Li, and Yeaple (2015). They study Chinese firms over 2001–06, within a firm, product and destination, with a closely related quality measure. Their coefficient on the change in a firm's input tariff, at $\sigma = 5$, is −3.093 on their measure. For the average cut of 5.98 percentage points this is a rise of $0.0598 \times 3.093 = 0.185$, and dividing by $\sigma - 1 = 4$ puts it on the scale of this paper: 0.046 in logs, about 4.7 percent. The two sizes are not scaled alike, since theirs is an average cut and mine compares the 10th and 90th percentiles of exposure.

Input-tariff cuts reduce the cost of supplying quality, and the baseline specification holds fixed input-cost changes common to a firm's US and non-US sales of a product (Section 3). More secure access to the US market instead changes which buyers firms can profitably serve.

## 5. Alternative explanations and robustness

This section asks whether exposure changes which relationships continue, whether the decline in quality is specific to the United States, and whether the mix of non-US destinations, quotas, cost or markup cuts, crowding, industry trends, exporter turnover or measurement choices can account for it. Each paragraph takes one question. Figure 4 summarizes the main 2002–04 robustness estimates, and Table A1 reports their exact values.

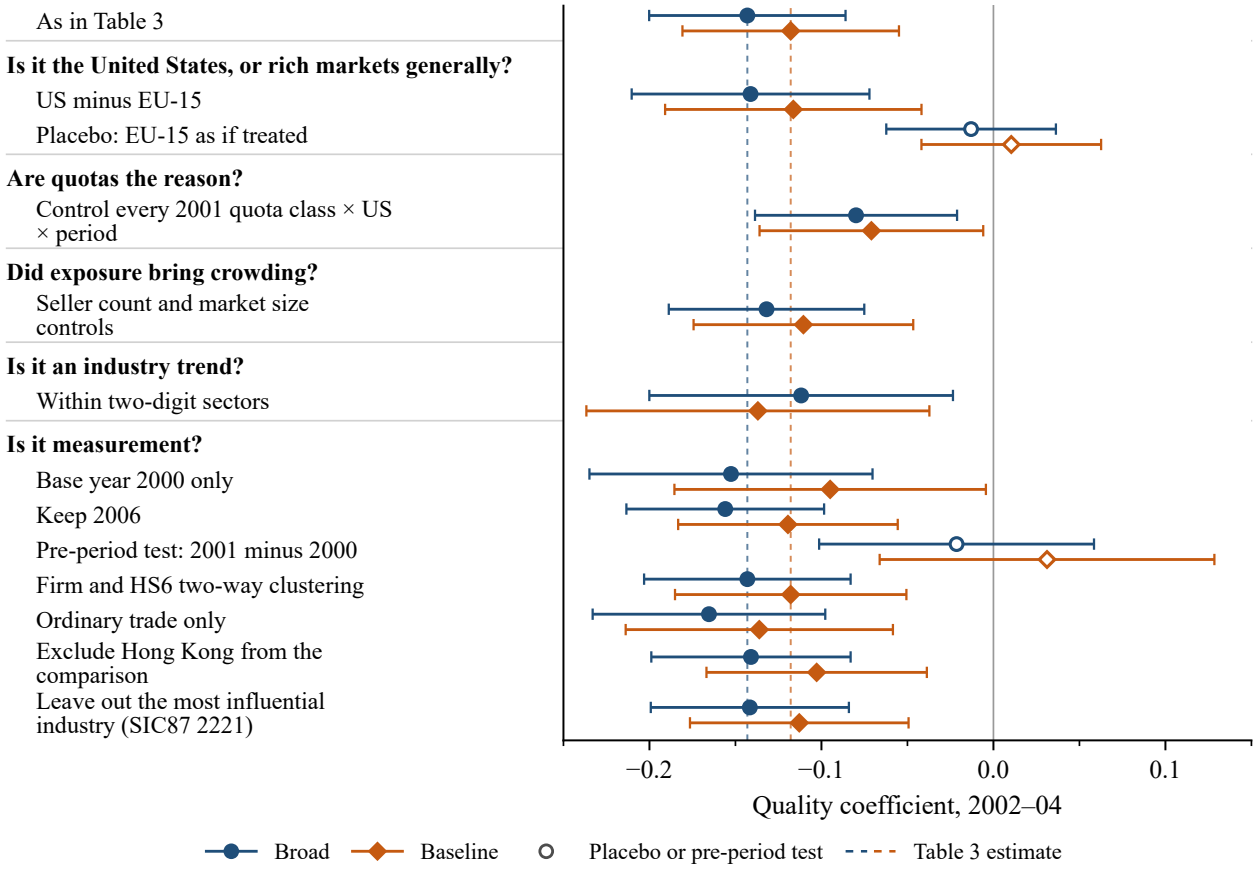



**Figure 4:** Selected robustness checks on the 2002–04 quality estimate

*Notes:* Each row is a selected 2002–04 robustness estimate of the quality coefficient with a 90 percent interval, for the broad and baseline specifications, grouped by the questions of this section; the dashed lines mark the Table 3 estimate in each specification's color. Hollow points are the placebo and the pre-period test, whose expected value is zero. The two-way clustering row has the point estimates of Table 3 with wider intervals. The base-year-2000 row is the mean of the 2002–04 event-study coefficients. The quota-class control interacts every product's 2001 quota class with the United States and with each period; whole 2001 quota categories are assigned to HS6 products, and products only partly covered by a limit form a class of their own. Exact numbers are in Table A1.

*Does exposure change who keeps selling?* The short-run changes are small. Table 5 shows that exposure moves the chance that a firm–product sells to both markets by +0.017 (standard error 0.010), and does not move the chance that a firm–product stops selling to the United States (−0.009, 0.026). Both change after 2005: selling to both rises (+0.140) and exit falls (−0.063). One margin does move in 2002–04: firm–products that did not sell to the United States before 2002 were less likely to start in exposed products (−0.067, 0.016). They are outside the

sample of continuing relationships and so cannot enter the estimate.

**Table 5:** PNTR exposure and selection into US export relationships

| Coefficient on NTR gap × US × period | 2002–04 | 2005–14 |
|---|---|---|
| Sells to both the US and elsewhere | +0.017* | +0.140*** |
| | (0.010) | (0.031) |
| Stops selling to the US | −0.009 | −0.063*** |
| | (0.026) | (0.018) |
| Starts selling to the US | −0.067*** | −0.019 |
| | (0.016) | (0.014) |

*Notes:* Each entry is the coefficient on NTR gap × US × period, with the standard error clustered by HS6 product in parentheses; 2006 dropped. The first row is a linear probability model on continuing relationships; the other two use the full panel. A relationship stops if it was shipped in 2000–01 and has no record in the window, and starts if it was not shipped in 2000–01 while the firm–product exported somewhere then. The stop row compares with the same firm–product's non-US sales; the start row has firm, product and destination effects, because a firm–product has only one relationship that could start. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

*Is it the United States, or rich markets generally?* Quality falls in the United States but not in the EU-15, which serves as the placebo region. The European Union granted China permanent most-favored-nation status in 1985, so China's WTO accession changed little in the tariff or the uncertainty its exporters faced there (Feng, Li, and Swenson, 2017). Treating the EU-15 as if it were treated, with US rows dropped, gives a 2002–04 quality coefficient of +0.010 (0.032). The difference between the United States and the EU-15 is −0.116 (0.045). Prices in the EU-15 also fell somewhat in exposed products, with a matching rise in quantity: the price coefficient is −0.049 (0.027) and the quantity coefficient +0.288 (0.098) (Table A6, Panel A).[10]

*Is it the changing mix of non-US destinations?* The customs extract distinguishes four blocks outside the United States: Japan, the EU-15, Hong Kong and all other destinations. Treating each block as a separate observation, so that the firm × product × year effect compares US sales with each block the firm serves, gives a 2002–04 estimate of −0.102 (0.038) (Table A2). Comparing the United States with one block at a time, the estimate is negative against each block, ranging from −0.102 against Japan to −0.157 against the EU-15, and is significant at the 5 percent level against every block except Japan (Table A2). The block of other destinations still pools many countries, and I do not split it further.

*Are quotas the reason?* US quotas on Chinese textiles and apparel changed during the sample, in 2002 and again in 2005, so their timing overlaps with PNTR. Controlling for each product's 2001 quota class, interacted with the United States and with each period, leaves the 2002–04 estimate at −0.071 (0.040), 60 percent of the estimate in Table 3, and significant at 10 percent (Figure 4 and Table A1).

[10]These coefficients compare the EU-15 with destinations outside the United States, Japan and Hong Kong. They come from a different regression from the placebo above, which treats the EU-15 as if treated and drops the US rows.

*Is it a markup or cost cut on an unchanged product?* Not on the evidence of prices and quantities. In Handley and Limão (2017) lower policy uncertainty leads exporters to upgrade their technology, which lowers marginal cost and with it the price. The firm × product × year effects of the baseline specification absorb any such cut that reaches the firm's product in all its markets. They do not absorb a markup or cost cut on US sales alone, but that cut leaves the product unchanged, so buyers move along their demand curve. Quantity would then have risen by far more than it did (Section 4.3), for any $\sigma$ above the threshold reported in Section 4.4.

*Did exposure bring crowding?* It brought more sellers, but controlling for the seller count and the size of the US market leaves the estimate close to where it was. Because both may themselves respond to PNTR, I read this as a diagnostic rather than as a decomposition. With these controls, the 2002–04 estimate is −0.110, 93 percent of the estimate in Table 3. Within a firm–product, relationships with more Chinese sellers tend to have lower quality: the coefficient on the seller count is −0.022 (0.013) (Table A6, Panel C). The two controls proxy for the market term that eq. (1) leaves out, which is common to all sellers of a product in a destination and year. The fixed effects cannot absorb that term, because it varies with exposure, the United States and the year, and the controls need not capture all of it.

*Is it an industry trend?* Within two-digit sectors the estimate is −0.137 (0.061), significant at 5 percent. Within four-digit sectors it is +0.020 and not significant (Table A1), but little exposure variation is left within those groups, and the estimate cannot be told apart from zero or from the two-digit estimate.

*Is it which firms export?* Table A4 splits the change in quality in each product's sales to each destination, from 2000–01 to 2002–04, into four terms following Melitz and Polanec (2015), as a post hoc description. Counted by sales, as Khandelwal, Schott, and Wei (2013) count, the product's quality does not move, and the fall in price that they trace to net entry does not appear here (entry plus exit is +0.032). New exporters look relatively better in exposed products' US sales, as in Feng, Li, and Swenson (2017) (+0.225, 0.076), though they already did between 2000 and 2001 (+0.093). The decline is in the typical continuing firm: −0.169 (0.082) when each survivor counts equally, which only a firm-by-firm count shows.

Feng, Li, and Swenson (2017) study entry and exit under PNTR, and Tables A3 and A5 put their design to my data. Entrants into exposed US products are cheaper, as they find (log price lower by 0.0060 per point of exposure in 2002, about 20 percent across the exposure range), but their quality conclusion does not hold on my data (−0.0067 at $\sigma = 5$, against their +0.009; positive only under raw product elasticities from Soderbery (2018)), and the within-firm decline does not depend on it. The price pattern is already there in 2001, before PNTR took effect (−0.0068), so this design does not separate PNTR from what came before it.

*Is it measurement?* The estimate stays negative under every measurement choice I report. Using only the year 2000 as the base (Table A1) gives −0.095 (0.055), significant at 10 percent. PNTR was signed into law in October 2000, so 2001 may be partly treated; to the extent that firms anticipated the policy, including 2001 in the base would bias the post-PNTR estimates toward zero. Keeping 2006 gives −0.119; restricting the sample to ordinary trade gives −0.136, suggesting that processing trade does not drive the result; excluding Hong Kong from the non-US comparison group, given its role as a re-exporter to the United States, gives −0.103; and omitting the

most influential SIC industry, a fabric mill, gives −0.113. Clustering jointly by firm and HS6 product leaves the point estimate unchanged and widens the confidence interval (standard error 0.041).

## 6. A model of selling down-market

Quality fell, and prices fell without the rise in quantity that an unchanged product would have brought (Section 4). This section offers a mechanism that produces both. When access to a market improves, a seller can profitably serve buyers it previously left alone. Those buyers value the attribute that quality stands for less than the buyers it already served, and serving them can lower the attribute the seller chooses. The model also shows that, for the sellers it describes, the quality index understates that fall. The environment, assumptions and proofs are in Appendices B and C.

Following Hottman, Redding, and Weinstein (2016), I call the attribute a seller chooses *appeal*: the characteristic of a variety that raises what a buyer will pay for it. Appeal is not the quality index of eq. (1), which moves with appeal and also with how much the seller sells. Section 6.4 relates the two.

### 6.1. Setup

Buyers differ in $\theta$, their marginal valuation of appeal, distributed on $[\underline{\theta}, \bar{\theta}]$. A seller chooses the appeal $a \geq 1$ and price of its variety, and which buyers to serve; each buyer relationship costs a flow $k$. Demand across sellers is CES with elasticity $\sigma$. Appeal raises marginal cost with elasticity $\eta$. The seller chooses appeal separately for each destination (Assumption 6), so the version sold to the United States can differ from the version sold elsewhere. The model has no entry or exit, as in the estimation sample of continuing relationships, and the price index of the seller's product group is held fixed (Assumption 1).

Access is the probability $\rho$ that the destination's tariff treatment is not reverted. I assume that PNTR raised $\rho$ to one (Assumption 2). Access enters the seller's problem as a weight $\varpi(\rho)$ on variable profit, which rises with $\rho$ for every product with a positive NTR gap.[11] The lowest buyer type served, the *cutoff* $\theta^*$, satisfies

$$\varpi(\rho)\, v(\theta^*; a) \;=\; k, \tag{3}$$

where $v(\theta; a)$ is the variable profit from a buyer of type $\theta$ at appeal $a$, which rises with $\theta$ (Lemma C.3). Holding appeal fixed, better access lowers the cutoff: the seller reaches further down.

### 6.2. The sorting boundary

Whether more appeal pays depends on whom the seller serves. With the seller serving buyers above the cutoff, the marginal value of appeal is

$$\Psi(S; a) = \int_S (\theta - \eta)\, \omega_a(\theta)\, d\theta, \qquad S = [\theta^*, \bar{\theta}], \tag{4}$$

where the weight $\omega_a(\theta) > 0$ rises with what buyer $\theta$ spends (Appendix B.1). A buyer whose valuation exceeds the cost elasticity $\eta$ raises the return to appeal; a buyer below $\eta$ lowers it. I call $\eta$ the *sorting boundary*. It is a

[11] For the 304 products with a zero gap $\varpi$ does not move, and the five products with a slightly negative gap are also exceptions (eq. (B.1)).

property of the cost function and is the same for every seller in the industry. The sorting follows Mussa and Rosen (1978), with constant-elasticity demand and a costly, endogenous reach in place of their monopolist's screening problem (Appendix B.1).

Figure 5 shows the case that matters. Before access improves, the seller serves only buyers above $\eta$. When the cutoff falls below $\eta$, the newly reached buyers between the new cutoff and $\eta$ value appeal less than it costs. Whether the overall return to appeal rises or falls depends on how many buyers the seller adds on each side of $\eta$, and it can go either way. The result below does not rely on that net change.

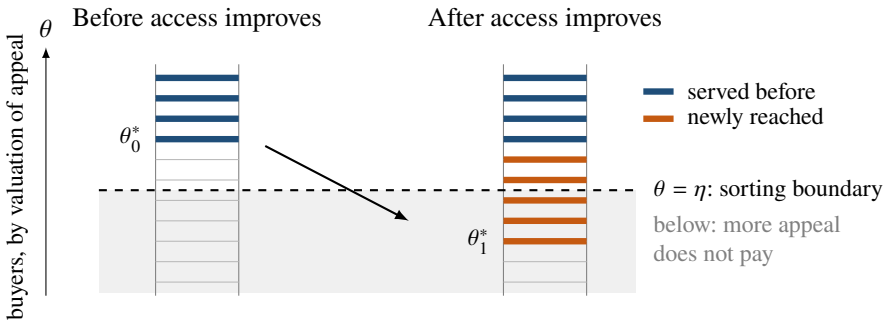



**Figure 5:** The sorting boundary and the cutoff

*Notes:* Schematic. Rungs are buyers ranked by their valuation of appeal $\theta$. Before access improves the seller serves buyers down to the cutoff $\theta_0^*$, all above the sorting boundary $\eta$. After, it serves buyers down to $\theta_1^* < \eta$; on the newly reached buyers below the dashed line, more appeal does not pay.

### 6.3. The reach cap

Reach and appeal are chosen together, so a lower return to appeal does not by itself imply lower appeal. The proposition instead bounds the appeal with which a buyer below $\eta$ can be profitably served. Write $T$ for the seller's *access index*, its own scale times the weight $\varpi(\rho)$, which is higher when access is better.

Proposition 6.1 (The reach cap). *Assume monotone scale (Assumption 3), the appeal normalisation (Assumption 4) and boundedness (Assumption 5), and that the seller offers one variety at each date (Definition C.2). If the variety serving a buyer $\tilde{\theta} < \eta$ has appeal $a$, then $a \leq \bar{a}(T, \tilde{\theta})$, where $\bar{a}$ is increasing in $T$ and decreasing in the distance of $\tilde{\theta}$ below $\eta$. If the variety's appeal before the access change exceeded $\bar{a}(T_1, \tilde{\theta})$, the cap at the new access index, its appeal falls.*

(Proof in Appendix C.) A buyer who values appeal less than it costs limits how much appeal can profitably be sold to that buyer. A seller whose appeal was set for buyers above $\eta$ and that now reaches a buyer far enough below $\eta$ must lower it. Better access raises the cap, but the fall occurs whenever the old appeal still exceeds the new cap. The cap comes from a feasibility comparison, not from first-order conditions, so it holds even where those conditions do not locate the optimum.

The condition is sufficient, not necessary, and it does not say that every seller downgrades. A seller whose cutoff stays above $\eta$, or whose appeal is already below the cap, faces no force lowering appeal. Because access, productivity and market size enter the seller's problem only through their product, which sellers reach below $\eta$ depends on where they started, and the trade data do not report that.

### 6.4. Appeal and the quality index

Equation (1) can be written $\ln \lambda = \ln p + \ln V/(\sigma - 1)$, where $V = pq$ is revenue. In the model the price is a constant markup over marginal cost, so $\ln p$ equals $\eta \ln a$ plus a term that does not depend on appeal (eq. (C.1)). The change in quality is therefore $\eta$ times the change in log appeal plus $1/(\sigma - 1)$ times the change in log revenue (Lemma C.7).

For a seller with one variety, revenue does not fall when access improves: the seller's value is linear in the access index, the index rose, and comparing the two optima shows that revenue cannot have fallen (Corollary C.9). Two things follow. Quality can fall only if appeal falls. And when it does fall, it falls by at most $\eta$ times the fall in log appeal (Corollary C.10), because the extra sales from wider reach enter the index as a rise. In this sense the quality index understates the fall in appeal. The converse needs a condition, since a fall in appeal can be offset by the revenue term; Proposition C.12 gives a sufficient one.

### 6.5. The estimates through the model

Each coefficient of Table 3 is a difference between the United States and other destinations, and the model describes each market separately (Assumption 6). I read the baseline 2002–04 estimates through it.

Quality fell, which the model produces through a fall in appeal by sellers whose reach crosses the sorting boundary (Proposition 6.1). The fall appears in 2002 and has not detectably reversed (Figure 2). The model compares two states of access, so it predicts a fall when access changes and says nothing about the year-by-year path.

Prices fell without a matching rise in quantity. In the model the log price moves by $\eta$ times the change in log appeal, and quantity responds both to the lower price and to the wider reach. The price and quantity coefficients sum to +0.059, the sign the model requires of a seller's revenue. The quality coefficient, −0.118, is accordingly smaller in absolute value than the price coefficient, −0.132; at $\sigma = 5$ the difference is that sum divided by four.[12] In Figure 3 the baseline point lies below the dashed line, so quality fell.

In the broad specification the price and quantity coefficients sum to −0.104. I derive no sign for that specification (Remark C.11); if the revenue restriction applied to it, the broad point estimates would violate it.

### 6.6. What the model does not identify

The trade data record a firm, a product, a destination group, a price and a quantity, but not the buyers. The model shows that this mechanism can produce the estimates; it does not show which buyers a firm reached. Three limits follow.

First, the baseline specification compares a firm's US and non-US shipments of the same product, and the model reads that comparison through appeal chosen separately for each destination. If one appeal served both markets, the baseline price coefficient would be exactly zero (Remark B.1). It is −0.132 and significant at the 1 percent level, so the product sold to the United States is not the same as the one sold elsewhere.

Second, a seller can also serve new buyers by adding a lower-appeal variety alongside the existing one, a possibility Manova and Zhang (2012) also raise. That too lowers the quality of what the relationship ships, and

[12] For a difference between destination groups the model implies these two signs only under the conditions of Remark C.11, which include that exposure did not raise non-US revenue more than US revenue. The non-US leg pools many markets, so its appeal is defined by its price, $\ln a_{\text{NUS}} \equiv (\ln p_{\text{NUS}} - \text{const})/\eta$.

customs records by firm and destination group cannot separate it from a fall in appeal.

Third, the model holds the product group's price index fixed (Assumption 1), so it is silent on entry by other sellers, which moves that index (Handley and Limão, 2017). Section 5 examines that channel.

## 7. Conclusion

Lower trade barriers can raise the quality of what firms export: when China cut its input tariffs, its exporters upgraded (Fan, Li, and Yeaple, 2015). This paper finds that a gain in market access moved quality the other way. When PNTR removed the threat that US tariffs on Chinese goods would revert to column-2 rates, continuing exporters lowered the quality of what they sold in the United States, by about 4.5 percent between the 10th and 90th percentiles of exposure. What changed is what existing exporters sell, not which firms export.

The explanation I propose is that access changes whom a firm sells to. Once the threat is gone, a seller can profitably reach buyers it previously left alone. If those buyers value quality less than the ones it already served, it pays to lower quality and sell down-market. The data do not identify those buyers. What the model shows is that this mechanism produces the pattern in the estimates: lower prices without the rise in quantity an unchanged product would bring.

The quality effect of liberalization therefore depends on what a reform changes. A reform that lowers the cost of supplying quality pushes firms up the quality ladder; one that widens the set of buyers they can reach can push them down it. Judging a trade reform by its effect on export quality requires knowing which of the two it does, and how much of each.

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## Appendix A. Additional results

This appendix gives the numbers behind the checks of Section 5, the post hoc decomposition of quality by exporter turnover, and the design of Feng, Li, and Swenson (2017) on my data. Unless noted, each table reports the 2002–04 coefficient on NTR gap × US × period for continuing relationships with 2006 dropped, quality at $\sigma = 5$, and the base period is the average of 2000 and 2001. Standard errors are clustered by HS6 product unless a note says otherwise. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

*The composition decomposition.* Table A4 is descriptive and was run after the main results. Following Melitz and Polanec (2015), for each product–destination the change in quantity-weighted mean log quality (or log price) between 2000–01 and 2002–04 equals the survivors' equal-weighted change, plus the change in the covariance between survivors' shares and quality, plus entrants against survivors, plus exiters against survivors; the identity holds exactly for every product–destination. I regress each component on exposure × US with product and destination effects. The covariance row splits exactly into the product's leading seller (the largest 2000–01 share; it holds 35 percent on average, against an equal share of 9 percent) against the rest, the size gradient among the rest, and sales moving at fixed quality.[13]

Counted by sales, as Khandelwal, Schott, and Wei (2013) count, the product's quality does not move: their within-firm term is +0.012, and entry, exit and sales moving add +0.010. Their price fall runs through net entry, and entry plus exit is +0.032 here. New exporters contribute more in exposed products' US sales, in the direction of Feng, Li, and Swenson (2017), but against exiters they are +0.128 on quality (standard error 0.192) and −0.002 on price, and the new-exporter term is already positive between 2000 and 2001. The decline is in the typical continuing firm, all but each product's leading seller, whose own change is imprecisely estimated. Only an equal-weighted count shows it. Weighted by the product–destination's 2000–01 export value, the total is +0.531 (standard error 0.221), and no component is significant.

*The design of Feng, Li, and Swenson (2017).* Table A5 reproduces their price fact: entrants into exposed US products are cheaper, and within two-digit sectors too through 2004. Their quality conclusion does not hold at $\sigma = 5$. The 2001 column is before PNTR took effect and the price pattern is already there, so this design does not separate PNTR from what came before it. All years are in Table A3.

The design follows theirs: firms with no exports in 2000 that export in year $t$ (entrants, measured in $t$), pooled with firm–products exported to a market in 2000 but not in $t$ (exiters, measured in 2000); the US against the EU-15, dropping US exporters from the EU sample (their footnote 39); HS6 × destination effects. My exposure is 100 times the 2000 gap; theirs subtracts the post-accession gap. Their data end in 2006.[14] The log price row is the interpretable one: −0.006 per point across the 10th-to-90th percentile of exposure (about 39 points) means prices for entrants about 20 percent lower.

[13] Firms adding the product (entrants minus new exporters: −0.170 for quality and −0.096 for price) have no separate standard error. The within-firm term of Khandelwal, Schott, and Wei (2013) is the Griliches–Regev term, weighted by each firm's average share and fitted on the 3,398 products that have both destinations in the pair. Because that is a different set of products, it does not add up with the four components.

[14] Their 2006 values are −4.080*** for price and +0.013*** for quality. They describe their price as the unit price over the product's median; on that measure mine is the first price row, and it is smaller and not significant before 2007. The magnitudes are not comparable: their price regressions have constants of 28.6–34.2 (their Table 8), far from the value near 1 that a ratio to the median implies, so their variable is either heavily right-skewed or scaled differently, and the paper does not say which.

**Table A1:** Robustness of the 2002–04 quality coefficient

| | Broad | Baseline |
|---|---|---|
| *Panel A. The checks in Figure 4* | | |
| As in Table 3 | −0.143*** | −0.118*** |
| | (0.035) | (0.038) |
| Seller count and market size controls | −0.132*** | −0.110*** |
| | (0.035) | (0.039) |
| Within two-digit sectors | −0.112** | −0.137** |
| | (0.054) | (0.061) |
| US minus EU-15 | −0.141*** | −0.116** |
| | (0.042) | (0.045) |
| Placebo: EU-15 as if treated | −0.013 | +0.010 |
| | (0.030) | (0.032) |
| Base year 2000 only | −0.153*** | −0.095* |
| | (0.050) | (0.055) |
| Keep 2006 | −0.156*** | −0.119*** |
| | (0.035) | (0.039) |
| Pre-period test: 2001 minus 2000 | −0.021 | +0.031 |
| | (0.049) | (0.059) |
| Firm and HS6 two-way clustering | −0.143*** | −0.118*** |
| | (0.036) | (0.041) |
| Control every 2001 quota class × US × period | −0.080** | −0.071* |
| | (0.036) | (0.040) |
| Ordinary trade only | −0.165*** | −0.136*** |
| | (0.041) | (0.047) |
| Exclude Hong Kong from the comparison | −0.141*** | −0.103*** |
| | (0.035) | (0.039) |
| Leave out the most influential industry (SIC87 2221) | −0.142*** | −0.113*** |
| | (0.035) | (0.039) |
| *Panel B. Further checks* | | |
| Within four-digit sectors | −0.052 | +0.020 |
| | (0.083) | (0.098) |
| Within two-digit sectors, base year 2000 only | −0.186** | −0.175** |
| | (0.076) | (0.085) |

*Notes:* Panel A gives the estimates plotted in Figure 4; Panel B adds two checks. The two-way clustering row clusters by firm and by HS6 product. The base-year-2000 rows are the mean of the 2002–04 event-study coefficients. The sector rows add sector × destination × year effects; little exposure variation is left within four-digit sectors. The Hong Kong row drops Hong Kong from the comparison group, and the leave-one-out row drops the SIC87 industry with the largest descriptive influence.

**Table A2:** Non-US destinations as separate observations

| | Broad | Baseline | Identifying firm–products |
|---|---|---|---|
| Pooled non-US group (main text) | −0.143*** | −0.118*** | 69,030 |
| | (0.035) | (0.038) | |
| Each non-US block a separate observation | −0.126*** | −0.102*** | 66,550 |
| | (0.035) | (0.038) | |
| United States against other destinations only | −0.151*** | −0.107** | 53,422 |
| | (0.039) | (0.044) | |
| United States against Japan only | −0.105** | −0.102 | 21,475 |
| | (0.050) | (0.065) | |
| United States against the EU-15 only | −0.188*** | −0.157*** | 43,044 |
| | (0.046) | (0.057) | |
| United States against Hong Kong only | −0.123** | −0.137** | 26,849 |
| | (0.050) | (0.063) | |

*Notes:* The destination blocks are the United States, Japan, the EU-15, Hong Kong and all other destinations. The first row pools the four non-US blocks into one group, as in the main text, and reproduces the main estimates. The second row keeps each non-US block as its own observation, with fixed effects by firm × product × block in place of firm × product × destination group. Each remaining row keeps only the United States and one block. The last column counts the firm–products that identify the baseline estimate (Section 3).

**Table A3:** Entrants against exiters in every year, on the design of Feng, Li, and Swenson (2017)

| | Entrant × US × exposure, by year | | | | | | |
|---|---|---|---|---|---|---|---|
| *Panel A. 2001–2007* | | | | | | | |
| | 2001 | 2002 | 2003 | 2004 | 2005 | 2006 | 2007 |
| FLS specification | | | | | | | |
| Price (log) | −0.0068*** | −0.0060*** | −0.0075*** | −0.0054*** | −0.0055*** | −0.0045*** | −0.0075*** |
| | (0.0022) | (0.0018) | (0.0017) | (0.0014) | (0.0014) | (0.0017) | (0.0019) |
| Quality, $\sigma = 5$ | −0.0078*** | −0.0067*** | −0.0077*** | −0.0062*** | −0.0059*** | −0.0051*** | −0.0074*** |
| | (0.0026) | (0.0020) | (0.0019) | (0.0016) | (0.0016) | (0.0019) | (0.0022) |
| Quality, Soderbery (2018), trimmed | −0.0079** | −0.0072*** | −0.0077*** | −0.0072*** | −0.0087*** | −0.0076*** | −0.0091*** |
| | (0.0032) | (0.0023) | (0.0022) | (0.0019) | (0.0020) | (0.0021) | (0.0021) |
| Quality, Soderbery (2018), raw | +0.0019 | +0.0016 | +0.0030* | +0.0015 | +0.0021 | +0.0013 | +0.0035** |
| | (0.0023) | (0.0018) | (0.0016) | (0.0014) | (0.0014) | (0.0014) | (0.0014) |
| Within two-digit sectors | | | | | | | |
| Price (log) | −0.0034 | −0.0042* | −0.0046** | −0.0041** | −0.0012 | −0.0017 | −0.0038 |
| | (0.0029) | (0.0024) | (0.0022) | (0.0020) | (0.0019) | (0.0022) | (0.0025) |
| Quality, $\sigma = 5$ | −0.0049 | −0.0048* | −0.0038 | −0.0039* | −0.0004 | −0.0016 | −0.0028 |
| | (0.0035) | (0.0028) | (0.0025) | (0.0022) | (0.0023) | (0.0024) | (0.0028) |
| Quality, Soderbery (2018), trimmed | −0.0072 | −0.0059* | −0.0019 | −0.0032 | +0.0001 | −0.0020 | −0.0021 |
| | (0.0045) | (0.0035) | (0.0031) | (0.0027) | (0.0027) | (0.0027) | (0.0027) |
| *Panel B. 2008–2014* | | | | | | | |
| | 2008 | 2009 | 2010 | 2011 | 2012 | 2013 | 2014 |
| FLS specification | | | | | | | |
| Price (log) | −0.0078*** | −0.0094*** | −0.0100*** | −0.0076*** | −0.0037** | −0.0002 | −0.0010 |
| | (0.0021) | (0.0020) | (0.0021) | (0.0020) | (0.0018) | (0.0021) | (0.0019) |
| Quality, $\sigma = 5$ | −0.0080*** | −0.0098*** | −0.0104*** | −0.0075*** | −0.0029 | +0.0012 | −0.0001 |
| | (0.0024) | (0.0023) | (0.0024) | (0.0023) | (0.0020) | (0.0023) | (0.0021) |
| Quality, Soderbery (2018), trimmed | −0.0114*** | −0.0129*** | −0.0149*** | −0.0120*** | −0.0089*** | −0.0052** | −0.0074*** |
| | (0.0021) | (0.0021) | (0.0022) | (0.0021) | (0.0021) | (0.0022) | (0.0020) |
| Quality, Soderbery (2018), raw | +0.0023 | +0.0027* | +0.0026* | +0.0026* | +0.0020 | +0.0025* | +0.0012 |
| | (0.0015) | (0.0014) | (0.0014) | (0.0014) | (0.0014) | (0.0014) | (0.0015) |
| Within two-digit sectors | | | | | | | |
| Price (log) | −0.0044 | −0.0042 | −0.0047* | −0.0020 | +0.0019 | +0.0027 | +0.0027 |
| | (0.0028) | (0.0027) | (0.0028) | (0.0026) | (0.0023) | (0.0025) | (0.0024) |
| Quality, $\sigma = 5$ | −0.0036 | −0.0034 | −0.0042 | −0.0004 | +0.0039 | +0.0052** | +0.0052** |
| | (0.0032) | (0.0031) | (0.0032) | (0.0029) | (0.0025) | (0.0026) | (0.0025) |
| Quality, Soderbery (2018), trimmed | −0.0026 | −0.0035 | −0.0045 | −0.0004 | +0.0031 | +0.0040 | +0.0040 |
| | (0.0029) | (0.0030) | (0.0030) | (0.0027) | (0.0027) | (0.0027) | (0.0025) |

*Notes:* Coefficient on entrant × US × exposure, per percentage point of exposure; entrants and exiters are defined in the text. FLS is Feng, Li, and Swenson (2017). The quality rows use $\sigma = 5$ or the product elasticities of Soderbery (2018), either as published ("raw") or winsorized at the 5th and 95th percentiles across products ("trimmed"). Estimates for 2006 are shown because it is the last year of Feng, Li, and Swenson (2017); the rest of the paper drops it.

**Table A4:** The composition margins in my design

| | Quality 2000–01 to 2002–04 | Price 2000–01 to 2002–04 | Quality 2000 to 2001 |
|---|---|---|---|
| Survivors' own change, each firm equal | −0.169** | −0.143** | −0.094 |
| | (0.082) | (0.070) | (0.089) |
| Bigger survivors against the rest | +0.178* | +0.086 | −0.096 |
| | (0.107) | (0.091) | (0.102) |
| leading seller against the rest | +0.126** | +0.123*** | — |
| | (0.052) | (0.046) | |
| larger non-leaders against smaller | +0.051 | −0.028 | — |
| | (0.041) | (0.034) | |
| sales moving at fixed quality | +0.001 | −0.009 | +0.008 |
| | (0.080) | (0.067) | (0.076) |
| Entrants against survivors | +0.055 | +0.081 | +0.303** |
| | (0.133) | (0.125) | (0.136) |
| new exporters (FLS's entrants) | +0.225*** | +0.178** | +0.093** |
| | (0.076) | (0.070) | (0.045) |
| Exiters against survivors | −0.046 | −0.049 | +0.158 |
| | (0.123) | (0.113) | (0.109) |
| Product's average | +0.018 | −0.024 | +0.270* |
| | (0.139) | (0.126) | (0.154) |
| *The two papers' own contrasts* | | | |
| KSW's within-firm term (share-weighted) | +0.012 | −0.010 | −0.110* |
| | (0.041) | (0.035) | (0.060) |
| New exporters against exiters (FLS) | +0.128 | −0.002 | +0.875*** |
| | (0.192) | (0.165) | (0.262) |

*Notes:* One observation per product–destination, unweighted, with product and destination effects: 5,630 product–destinations in 2,815 products with survivors in both destinations. Shares are quantity shares, quality is at $\sigma = 5$, and 2006 is not used. Multiply by 0.37 to read the 10th-to-90th percentile difference in exposure on these products. KSW is Khandelwal, Schott, and Wei (2013) and FLS is Feng, Li, and Swenson (2017). Dashes mark rows not computed for that period.

Their quality measure needs a demand elasticity for each product, and on my data it exists for only about a sixth of products. They estimate $\sigma$ product by product, regressing log quantity on log price with HS6 × year effects. On Chinese exports to the United States and the EU-15 for 2000–2006 this gives an estimate for 3,914 HS6 products, of which 651 exceed 1. The median estimate is 0.58 and the 90th percentile is 1.19. Quality is undefined for $\sigma \leq 1$, so any estimate built on their measure rests on that sixth of products.[15]

**Table A5:** Entrants against exiters on the design of Feng, Li, and Swenson (2017)

| Entrant × US × exposure | $t$ = 2001 (pre) | $t$ = 2002 | $t$ = 2004 | $t$ = 2005 | $t$ = 2007 |
|---|---|---|---|---|---|
| *FLS (2017), Table 8, published* | | | | | |
| Price | — | −4.187* | −4.836*** | — | — |
| | | (2.308) | (1.374) | | |
| Quality | — | +0.009 | +0.003 | — | — |
| | | (0.007) | (0.005) | | |
| *Ours, FLS specification* | | | | | |
| Price relative to product median | −0.0997 | −0.0804 | −0.3455 | −0.1508 | −0.2244** |
| (FLS's measure) | (0.1072) | (0.1362) | (0.3471) | (0.1812) | (0.1097) |
| Price (log unit value) | −0.0068*** | −0.0060*** | −0.0054*** | −0.0055*** | −0.0075*** |
| | (0.0022) | (0.0018) | (0.0014) | (0.0014) | (0.0019) |
| Quality | −0.0078*** | −0.0067*** | −0.0062*** | −0.0059*** | −0.0074*** |
| | (0.0026) | (0.0020) | (0.0016) | (0.0016) | (0.0022) |
| *Ours, within two-digit sectors* | | | | | |
| Price (log unit value) | −0.0034 | −0.0042* | −0.0041** | −0.0012 | −0.0038 |
| | (0.0029) | (0.0024) | (0.0020) | (0.0019) | (0.0025) |
| Quality | −0.0049 | −0.0048* | −0.0039* | −0.0004 | −0.0028 |
| | (0.0035) | (0.0028) | (0.0022) | (0.0023) | (0.0028) |
| *Entrant premium (entrant indicator), ours* | | | | | |
| Quality | +0.0011 | +0.0449 | +0.2058*** | +0.3569*** | +0.6826*** |
| | (0.0812) | (0.0668) | (0.0540) | (0.0505) | (0.0556) |

*Notes:* Coefficients per percentage point of exposure. The 2001 column is before PNTR took effect. Quality is at $\sigma = 5$; other values of $\sigma$ are in Table A3. The last row is the coefficient on the entrant indicator alone: the quality of entrants relative to exiters in EU-15 sales of a product with zero exposure. The published coefficients are from Table 8 of Feng, Li, and Swenson (2017), with their significance stars.

[15] Many estimates probably fall below 1 because prices respond to demand shocks, which biases least squares demand regressions towards zero.

**Table A6:** More checks: EU-15 prices and quantities, crowding, and sectors in 2005–14

| | Broad | Baseline |
|---|---|---|
| *Panel A. EU-15 against destinations outside the US, Japan and Hong Kong, 2002–04* | | |
| Quality | −0.006 | +0.010 |
| | (0.032) | (0.033) |
| Unit value | −0.048* | −0.049* |
| | (0.026) | (0.027) |
| Quantity | +0.216*** | +0.288*** |
| | (0.077) | (0.098) |
| *Panel B. Quality within two-digit sectors, by period* | | |
| 2002–04 | −0.112** | −0.137** |
| | (0.054) | (0.061) |
| 2005–14 | −0.079 | −0.109 |
| | (0.091) | (0.092) |
| 2005–14 minus 2002–04 | +0.033 | +0.028 |
| | (0.087) | (0.090) |
| *Panel C. Crowding: quality regression with seller-count and US market-size controls* | | |
| Log number of Chinese sellers | −0.034*** | −0.022* |
| | (0.013) | (0.013) |

*Notes:* Panel A reports the 2002–04 coefficient on exposure × EU-15 × period for quality, unit value and quantity; the comparison group is destinations outside the United States, Japan and Hong Kong. Panel B repeats the quality estimate within two-digit HS chapters for 2002–04 and 2005–14 and gives the difference. Panel C reports the coefficient on the log number of Chinese sellers in the quality regression that adds the seller-count and market-size controls.

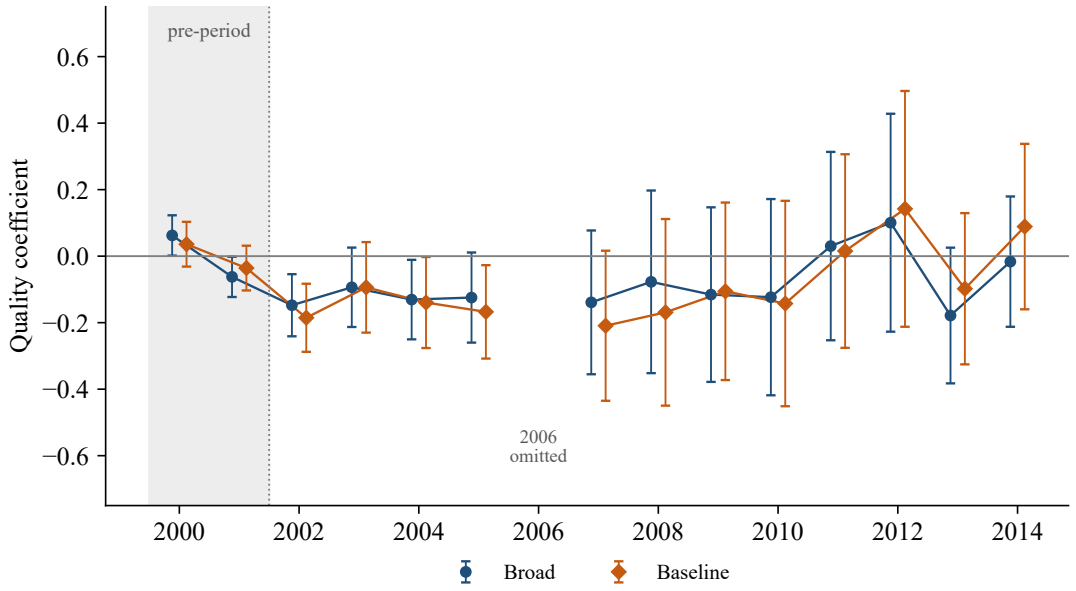



**Figure A1:** Quality by year with two-digit sector effects

*Notes:* Figure 2 re-estimated with two-digit sector × destination × year effects: coefficients on exposure × US by year, broad and baseline specifications, with 90 percent intervals.

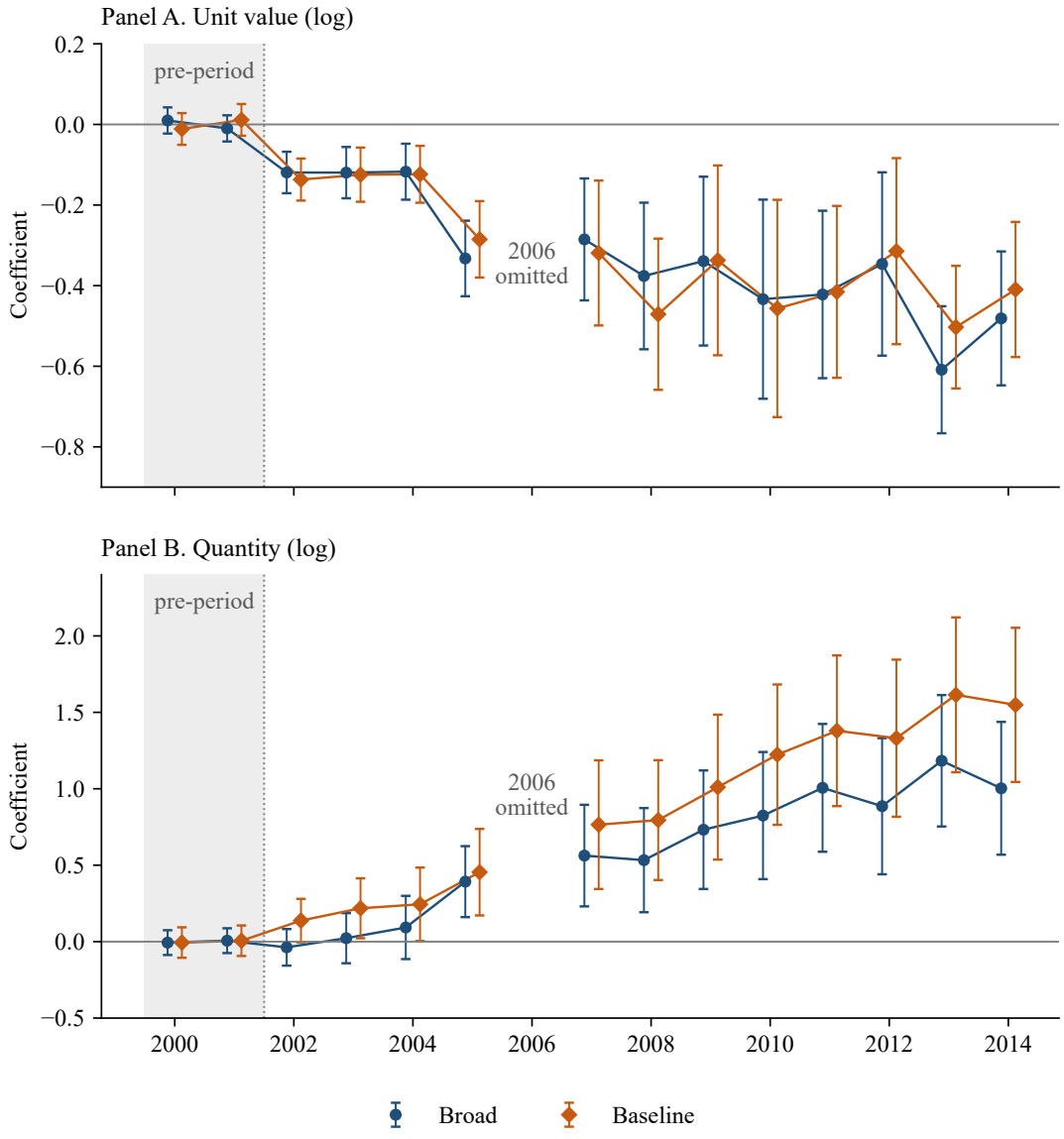



**Figure A2:** Unit value and quantity by year

*Notes:* The event study of Figure 2 with log unit value (Panel A) and log quantity (Panel B) as outcomes, broad and baseline specifications, with 90 percent intervals.

## Appendix B. The model: environment and assumptions

This appendix sets out the environment of Section 6 in full and states the assumptions. Appendix C contains the proofs. Table A7 collects the notation.

### B.1. Environment

*Ingredients.* The industry side, a continuum of sellers of heterogeneous productivity facing CES demand and paying a fixed cost per product, follows Melitz (2003). Buyers of different types sorting across qualities follows Fajgelbaum, Grossman, and Helpman (2011). That appeal raises marginal cost is the regularity documented by Kugler and Verhoogen (2012), where more capable plants buy higher-appeal inputs, pay more for them, and sell higher-appeal output at higher prices. The new element is that reach and appeal are chosen jointly and are linked through one elasticity.

The sorting is that of Mussa and Rosen (1978): buyers differ in what they will pay for appeal, and marginal cost rises with appeal. The environment differs from theirs in three ways. Demand across sellers is CES, so a buyer's quantity responds to price. The buyer's type enters as an exponent on appeal, $a^{(\sigma-1)(\theta-\eta)}$, rather than as a level of willingness to pay. And each buyer relationship costs a flow, which makes reach a choice. The force also differs: in Mussa and Rosen the seller distorts appeal downward to screen a fixed population of buyers, whereas here the return to appeal changes because the population served changes.

*Buyers.* A buyer of type $\theta$ has outlay $Y(\theta)$, of which a share $\psi(\theta)$ is spent on the seller's product group, whose price index is $P(\theta)$. The three enter only through

$$m(\theta) \;\equiv\; Y(\theta)\,\psi(\theta)\,P(\theta)^{\sigma-1} \;>\; 0.$$

With $g$ the density of $\theta$, the weight a served buyer carries in the seller's choice of appeal is proportional to

$$\omega_a(\theta) \;\propto\; m(\theta)\,a^{(\sigma-1)\theta}\,g(\theta).$$

The exact weight, (C.6), multiplies this by factors that do not depend on $\theta$, so every sign statement made from it is unaffected.

*Timing.* Nature draws productivities and the probability $\rho$ that the destination's tariff treatment is not reverted. Each seller then chooses, at the same time, which buyers to serve and the appeal of its variety, taking price indices and rivals' offers as given. Reach and appeal are chosen together because each determines the return to the other: the buyers served determine the return to appeal, and appeal determines which buyers are worth serving. The tariff state is then realised. Buyers allocate spending across sellers and, within a seller, buy the variety that is cheapest in appeal-adjusted terms. Because the flow cost of a relationship is paid in either tariff state, the seller ships to every buyer it maintains in both states, so the set of buyers served is the same in both and there is no second cutoff in the reverted state.

*Access.* In the reverted state the environment is the same at a higher tariff, so the value of a relationship is a fixed fraction $\chi$ of its value in the good state. With $t_1$ the NTR tariff and $t_2$ the column-2 rate, $\chi = [(1+t_2)/(1+t_1)]^{-\sigma}$, which depends on neither the buyer's type nor appeal and lies in $(0, 1)$ when $t_2 > t_1$. The policy therefore enters every margin of the seller's problem through the single weight

$$\varpi(\rho) \equiv \rho + (1-\rho)\,\chi, \qquad \varpi(0) = \chi, \quad \varpi(1) = 1, \tag{B.1}$$

which is strictly increasing in $\rho$ when $\chi < 1$. For the five products in the estimation sample with a slightly negative gap, $\chi$ is slightly above one and PNTR lowers the weight.

Modelling the policy as a probability that tariff treatment is not reverted follows Handley and Limão (2017). In their model uncertainty works through sunk entry costs and the option value of waiting. Here reach costs a flow, so there is no option value and the policy reduces to the weight $\varpi$. The two margins are complementary: theirs governs whether a seller enters the market, this one how far down it sells once it is there. Because $\chi$ is a ratio of tariff factors, $\varpi$ differs across products with different schedules.

*Reach regimes.* Access, productivity and the size of the market enter the seller's problem only through their product, the access index $T$ of (C.2). A seller with twice the productivity and one with twice the access weight solve the same problem, so the model alone cannot tell better access from more productive sellers. As $T$ rises, a seller passes from serving only buyers above $\eta$ to serving some buyers below it (Corollary C.4), at a threshold that does not depend on appeal. Better access multiplies $T$ by a common factor for every seller of a product, larger the larger the gap. Which sellers cross the threshold depends on where they started, and the customs data do not reveal that.

### B.2. Assumptions

The assumptions below restrict buyer spending, the price index, the policy and the cost of appeal. None can be tested with customs data, which record neither buyers nor their valuations.

Assumption 1 (Fixed group price index (CLOSURE)). *Within a date, all buyers face one price index for the seller's product group: $P_g(\theta) \equiv P_{g,t}$, constant across $\theta$.*

As in Melitz (2003), each seller takes the price index as given. The level of the index is absorbed into the fixed cost; its change between the two dates is held fixed by Definition C.2(i). The results are therefore partial equilibrium in the price index and say nothing about entry by other sellers, which moves it (Section 6.6).

Assumption 2 (PNTR eliminates the threat of reversion (ELIM)). *Before PNTR every product faces the same probability $1 - \rho_0 > 0$ that the column-2 schedule is reinstated. PNTR sets $\rho_1 = 1$ for every product.*

This is the case in Handley and Limão (2017) in which accession eliminated uncertainty and all industries face the same policy state; their alternative, in which a positive probability of a policy change remains after accession, is excluded. With $\rho_1 = 1$ the post-PNTR access index does not depend on exposure, which Remark C.11 uses.

Assumption 3 (Monotone spending (MON)). *Write $\bar{m}(\theta) > 0$ for buyer $\theta$'s spending on the seller's product group, net of the seller's own scale. Then $d \ln \bar{m}(\theta)/d\theta > 0$.*

In Verhoogen (2008) and Fajgelbaum, Grossman, and Helpman (2011), willingness to pay for quality rises with income; Assumption 3 assumes the corresponding ordering of spending, so that buyers who value appeal more also spend more on the group. This delivers a set of buyers served of the form $[\theta^*, \bar{\theta}]$ and hence the single cutoff used throughout (Lemma C.3). Under Assumption 1 it is a condition on outlay times the share spent on the group.

Assumption 4 (Appeal normalisation (NORM)). $a \geq 1$ *for every variety.*

Assumption 5 (Boundedness (BND)). $\kappa > (\sigma - 1)(\bar{\theta} - \eta)$*: the elasticity of the fixed cost in appeal exceeds the largest elasticity of variable profit in appeal.*

This bounds the seller's objective so that a maximiser exists (Lemma C.1). It is the only place it is used.

Assumption 6 (Destination-specific appeal (MKT)). *The seller chooses an appeal $a_d$ for each destination market, and the results of Section 6 and Appendix C hold market by market.*

Verhoogen (2008) models a plant that chooses quality separately for each market it serves, and Manova and Zhang (2012) find in Chinese customs data that firms customize a product's quality across destinations. No proof uses this assumption; the seller's problem is already stated at a destination-specific access index. It allows the results to be read across the comparison of Section 3, as the difference of two statements each made within one market. The alternative, one appeal for both markets, is rejected by the estimates.

Remark B.1 (Common appeal is rejected by the estimates). Step 1. *By* (C.1) *the price is* $p(a) = \frac{\sigma}{\sigma-1}(c_0/\varphi)\, a^{\eta}$*, with no destination index. If the same $a$ served both markets at each date, the price would be the same in both at each date. In the baseline specification, whose firm × product × year effect absorbs any term common to the two legs, the price coefficient of eq.* (2) *would then be exactly zero. (The broad specification has no such effect and includes firm–products observed on one leg only, so common appeal need not set its price coefficient to zero.) The baseline 2002–04 price coefficient is* $-0.132$ *(*$0.031$*) (Table 3, Panel B).*

Step 2. *Since* $\ln \lambda = \frac{\sigma}{\sigma-1} \ln p + \frac{1}{\sigma-1} \ln q$*, a zero price coefficient would make the baseline quality coefficient* $1/(\sigma - 1)$ *times the quantity coefficient:* $+0.048$ *at* $\sigma = 5$*, against the estimated* $-0.118$ *(*$0.038$*) (Table 3, Panels A and C).*

*What is rejected is common appeal together with constant-markup pricing* (C.1) *and Definition C.2. The latter two are part of the model, so I read the rejection as falling on common appeal: the version sold to the United States is not the same as the one sold elsewhere, as Assumption 6 states. The argument is that of Manova and Zhang (2012): with constant markups, price differences across destinations within a firm–product reflect differences in quality.*

## Appendix C. Proofs

Appendix C.1 writes down the seller's problem, Appendix C.2 proves Proposition 6.1, and Appendix C.3 relates chosen appeal to the quality index.

**Table A7:** Notation

| Symbol | Meaning |
|---|---|
| *Primitives* | |
| $\theta \in \Theta = [\underline{\theta}, \bar{\theta}]$ | A buyer's marginal valuation of appeal, with density $g > 0$ |
| $\eta > 0$ | Elasticity of marginal cost in appeal; the sorting boundary |
| $\sigma > 1$ | Elasticity of substitution across sellers |
| $a \geq 1$ | Appeal of a variety |
| $k$ | Flow cost of one buyer relationship |
| $F_0 a^\kappa$ | Fixed cost of a variety |
| $\varphi$, $x \equiv \varphi^{\sigma-1}$ | Seller productivity and scale |
| $\rho$ | Probability that the destination's tariff treatment is not reverted |
| $t_1, t_2$ | NTR and column-2 tariff rates |
| *Choices* | |
| $\theta^*$ | Cutoff: the seller serves $[\theta^*, \bar{\theta}]$; $\theta_0^*$, $\theta_1^*$ before and after the access change |
| *Derived objects* | |
| $\bar{m}(\theta)$ | Buyer $\theta$'s spending on the group, net of seller scale |
| $\chi$, $\varpi(\rho)$ | Reverted-state value fraction; access weight on variable profit, (B.1) |
| $T \equiv c\, \varpi(\rho)\, x$ | Access index, (C.2) |
| $\bar{\Gamma}(a; S)$ | Appeal-weighted value of the buyers in $S$, (C.4) |
| $\mathsf{N}(\theta^*)$ | Mass of buyers served |
| $\pi(a, \theta^*; T)$ | Seller's value, (C.3) |
| $V$ | Revenue, (C.5) |
| $\Psi(S; a)$, $\omega_a(\theta)$ | Marginal value of appeal when serving $S$, and a buyer's weight in it, (4) |
| $v(\theta; a)$ | Good-state variable profit from a buyer of type $\theta$ at appeal $a$ |
| $\iota(\theta; a, T)$ | Participation index of buyer $\theta$, (C.7) |
| $\bar{a}(T, \tilde{\theta})$ | Reach cap that a buyer $\tilde{\theta} < \eta$ places on appeal, (C.10) |
| *Constants* | |
| $c$, $\bar{c}$ | Scale constants in the access index, revenue and (C.4) |
| $\tau$ | Date- and destination-specific factors outside the seller's choice, (C.5) |

### C.1. The seller's problem

A seller with scale $x \equiv \varphi^{\sigma-1}$ chooses an appeal $a \geq 1$ and a set of buyers to serve. Marginal cost is $c_0 a^{\eta}/\varphi$, and CES demand with elasticity $\sigma > 1$ gives the constant-markup price

$$p(a) \;=\; \frac{\sigma}{\sigma-1}\,\frac{c_0}{\varphi}\,a^{\eta}, \qquad \text{so} \qquad p(a)\,a^{-\eta} \text{ does not depend on } a. \tag{C.1}$$

A relationship with a buyer costs the flow $k$, and a variety costs the fixed amount $F_0 a^{\kappa}$. Write $\bar{m}$ for buyer spending net of the seller's scale, $g$ for the density of types on $\Theta = [\underline{\theta}, \bar{\theta}]$, and

$$T \;\equiv\; c\,\varpi(\rho)\,x \tag{C.2}$$

for the access index, which is strictly increasing in $\rho$ for a product with $t_2 > t_1$. With one variety, the seller's value from serving $[\theta^*, \bar{\theta}]$ is

$$\pi(a, \theta^*; T) \;=\; T\,\bar{\Gamma}\big(a; [\theta^*, \bar{\theta}]\big) \;-\; F_0 a^{\kappa} \;-\; k\,\mathsf{N}(\theta^*), \tag{C.3}$$

where

$$\bar{\Gamma}(a; S) \;\equiv\; \int_S \bar{c}\,\bar{m}(\theta)\,a^{(\sigma-1)(\theta-\eta)}\,g(\theta)\,d\theta, \qquad \mathsf{N}(\theta^*) \;\equiv\; \int_{\theta^*}^{\bar{\theta}} g(\theta)\,d\theta. \tag{C.4}$$

The value is affine in $T$, so the policy enters only as a weight on variable profit. Revenue in the good state is

$$V \;=\; \sigma\,c\,x\,\tau\,\bar{\Gamma}\big(a; [\theta^*, \bar{\theta}]\big), \tag{C.5}$$

where $\tau$ collects date- and destination-specific factors outside the seller's choice. The weight $\varpi$ does not appear in (C.5), because it weights expected value across the two tariff states, not revenue in the good state.

The proofs use Assumption 3 (MON), Assumption 4 (NORM) and Assumption 5 (BND).

Lemma C.1 (A maximiser exists). *Assume* (MON), (NORM) *and* (BND). *Then* $\pi(\cdot, \cdot; T)$ *of* (C.3) *attains a maximum on* $[1, \infty) \times \Theta$.

Proof. By (MON) $\bar{m}$ is increasing on $\Theta$ and finite at $\bar{\theta}$, so $\bar{m} \leq \bar{m}(\bar{\theta}) < \infty$ there, and $a^{(\sigma-1)(\theta-\eta)} \leq a^{(\sigma-1)(\bar{\theta}-\eta)}$ for every $a \geq 1$ and $\theta \in \Theta$. Hence $\bar{\Gamma}(a; S) \leq \bar{c}\,\bar{m}(\bar{\theta})\,a^{(\sigma-1)(\bar{\theta}-\eta)}$ for every $S \subseteq \Theta$, so the first term of (C.3) grows at most like $a^{(\sigma-1)(\bar{\theta}-\eta)}$ while the second falls like $-F_0 a^{\kappa}$ with $\kappa > (\sigma-1)(\bar{\theta}-\eta)$ by (BND). Therefore $\pi(a, \theta^*; T) \to -\infty$ as $a \to \infty$, uniformly in $\theta^*$, and there is $A < \infty$ with $\pi(a, \theta^*; T) < \pi(1, \bar{\theta}; T)$ for all $a > A$ and all $\theta^*$. On the compact $[1, A] \times \Theta$ the map $(a, \theta^*) \mapsto \pi$ is continuous, since $\bar{\Gamma}$ and $\mathsf{N}$ are integrals of an integrable function over an interval with a moving endpoint. A continuous function on a compact set attains its maximum, and by the bound above that maximum is a maximum on the whole of $[1, \infty) \times \Theta$. □

None of the results below uses a first-order condition, a second-order condition, or a single-crossing or concavity restriction on $a$. They therefore hold even where first-order conditions do not locate the optimum.

Definition C.2 (The two-date configuration (CONF)). *A seller's pair of dates satisfies* (CONF) *if:* (i) *the buyer-side schedules* $(P, \psi)$ *are the same at the two dates, so that* $\bar{m}$ *and* $g$ *do not move;* (ii) *exactly one variety is*

*active at each date;* (iii) *the factors collected in* $\tau$ *are the same at the two dates; and* (iv) *the access change raises the index,* $T_1 > T_0$.

Differentiating (C.4) under the integral, on any $S$ with $\bar{m}g$ integrable,

$$T\,\frac{\partial \bar{\Gamma}(a;S)}{\partial \ln a} \;=\; \int_S (\theta-\eta)\,\omega_a(\theta)\,d\theta, \qquad \omega_a(\theta) \;\equiv\; (\sigma-1)\,T\,\bar{c}\,\bar{m}(\theta)\,a^{(\sigma-1)(\theta-\eta)}g(\theta) \;>\; 0, \tag{C.6}$$

which is eq. (4). Buyers above $\eta$ raise the marginal value of appeal and buyers below it lower it.

### C.2. The reach cap

Define the *participation index* of buyer $\theta$ at appeal $a$ and access index $T$,

$$\iota(\theta;a,T) \;\equiv\; T\,\bar{c}\,\bar{m}(\theta)\,a^{(\sigma-1)(\theta-\eta)}, \tag{C.7}$$

so that from (C.3) and (C.4),

$$\frac{\partial \pi}{\partial \theta^*} \;=\; g(\theta^*)\,\big[\,k-\iota(\theta^*;a,T)\,\big]. \tag{C.8}$$

Lemma C.3 (Served buyers cover the flow cost). *Assume* (MON) *and* (NORM). *Then*

(i) $\theta \mapsto \iota(\theta;a,T)$ *is strictly increasing on* $\Theta$*, so the set* $\{\theta : \iota(\theta;a,T) \geq k\}$ *is an upper interval and the seller serves* $[\theta^*,\bar{\theta}]$ *for some* $\theta^*$*;*

(ii) *at any maximiser* $(a,\theta^*)$ *of* (C.3)*,* $\iota(\theta^*;a,T) \geq k$*, with equality whenever* $\theta^* > \underline{\theta}$*; consequently* $\iota(\theta;a,T) \geq k$ *for* every *served* $\theta$.

Proof. *Step 1 (part (i)).* From (C.7), $d\ln\iota/d\theta = d\ln\bar{m}(\theta)/d\theta + (\sigma-1)\ln a$. The first term is strictly positive by (MON) and the second is non-negative by (NORM), so $\iota$ is strictly increasing in $\theta$. A strictly increasing function exceeds a threshold on an upper interval, and the seller serves $\theta$ exactly when the relationship pays its flow cost, that is, when $\iota(\theta;a,T) \geq k$.

*Step 2 (part (ii), interior cutoff).* If $\theta^* > \underline{\theta}$ is optimal then $\partial\pi/\partial\theta^* = 0$ there, and by (C.8) with $g > 0$ this is $\iota(\theta^*;a,T) = k$.

*Step 3 (part (ii), corner cutoff).* Suppose $\theta^* = \underline{\theta}$ and $\iota(\underline{\theta};a,T) < k$. Then (C.8) gives $\partial\pi/\partial\theta^* = g(\underline{\theta})[k - \iota(\underline{\theta};a,T)] > 0$, so raising $\theta^*$ above $\underline{\theta}$, that is, dropping the marginal buyers, strictly raises $\pi$, contradicting optimality. Hence $\iota(\underline{\theta};a,T) \geq k$. This step compares values and uses no derivative in $a$.

*Step 4.* By Steps 2–3 the index is at least $k$ at the cutoff; by Step 1 it is increasing in $\theta$; so it is at least $k$ at every $\theta \geq \theta^*$. □

Corollary C.4 (Who reaches below the sorting boundary). *Assume* (MON) *and* (NORM). *The boundary buyer* $\eta$ *is served if and only if*

$$T \;\geq\; \frac{k}{\bar{c}\,\bar{m}(\eta)}, \tag{C.9}$$

*and if the inequality is strict and* $\bar{m}$ *is continuous at* $\eta$ *then some buyer strictly below* $\eta$ *is served, so Proposition 6.1*

*applies. The threshold does not involve appeal.*

PROOF. By (C.7), $\iota(\eta; a, T) = T\bar{c}\,\bar{m}(\eta)$, because the appeal term is $a^0 = 1$ at $\theta = \eta$. By Lemma C.3(i) a buyer is served exactly when $\iota \geq k$, which at $\eta$ is (C.9). If the inequality is strict and $\bar{m}$ is continuous at $\eta$, then $\iota > k$ on a neighbourhood of $\eta$, and $\iota$ is strictly increasing, so the seller serves buyers strictly below $\eta$. □

Whether a seller reaches below the sorting boundary therefore depends on its access index and on primitives at $\eta$, not on the appeal it chooses. The corollary does not sign the resulting change in appeal; that requires the condition of Proposition 6.1. The threshold involves $k$, $\bar{c}$ and $\bar{m}(\eta)$, none of which is observed. Initial revenue ranks sellers' access indices within a product, destination and date (eq. (C.5)) but does not locate the threshold, so splitting sellers by initial revenue cannot test the mechanism.

LEMMA C.5 (The reach cap). *Assume* (MON) *and* (NORM)*, and let* $(a, \theta^*)$ *be any maximiser of* (C.3) *at access index* $T$*. If the maximiser serves a buyer* $\tilde{\theta} < \eta$*, then*

$$a \;\leq\; \bar{a}(T, \tilde{\theta}) \;\equiv\; \left(\frac{T\,\bar{c}\,\bar{m}(\tilde{\theta})}{k}\right)^{\frac{1}{(\sigma-1)(\eta-\tilde{\theta})}}, \tag{C.10}$$

*and* $\bar{a}(T, \tilde{\theta}) \geq 1$*. Moreover, writing* $d \equiv \eta - \tilde{\theta} > 0$ *for the distance of the buyer below* $\eta$*,* $\bar{a}$ *is*

(i) *strictly increasing in* $T$*, hence, for a product with* $t_2 > t_1$*, strictly increasing in* $\rho$ *along* (C.2)*; and*

(ii) *strictly decreasing in* $d$ *on the range of* $d$ *at which* $\tilde{\theta}$ *can be served at all.*

PROOF. *Step 1 (the cap).* By Lemma C.3(ii) the served buyer $\tilde{\theta}$ satisfies $T\bar{c}\,\bar{m}(\tilde{\theta})\,a^{(\sigma-1)(\tilde{\theta}-\eta)} \geq k$, that is,

$$a^{-(\sigma-1)(\eta-\tilde{\theta})} \;\geq\; \frac{k}{T\,\bar{c}\,\bar{m}(\tilde{\theta})}. \tag{C.11}$$

Because $\tilde{\theta} < \eta$ the exponent on the left is strictly negative, so raising both sides to the power $-1/[(\sigma-1)(\eta-\tilde{\theta})] < 0$ reverses the inequality and gives (C.10).

*Step 2 ($\bar{a} \geq 1$).* By (NORM), $a \geq 1$, so the left side of (C.11) is at most 1; hence $T\bar{c}\,\bar{m}(\tilde{\theta}) \geq k$, and $\bar{a}(T, \tilde{\theta}) \geq 1$. A buyer below $\eta$ with $T\bar{c}\,\bar{m}(\tilde{\theta}) < k$ cannot be made worth serving at any admissible appeal and is not served.

*Step 3 (monotonicity in $T$).* $\ln \bar{a} = \ln\big(T\bar{c}\bar{m}(\tilde{\theta})/k\big)\,\big/\,[(\sigma-1)d]$ is strictly increasing in $T$ at fixed $d$, because the numerator is and the denominator is positive and constant. $T$ is strictly increasing in $\rho$ for a product with $t_2 > t_1$, by (C.2) and the monotonicity of $\varpi$.

*Step 4 (monotonicity in $d$).* Write $N(d) \equiv \ln\big(T\bar{c}\,\bar{m}(\eta - d)/k\big)$, so that $\ln \bar{a} = N(d)/[(\sigma-1)d]$. By Step 2, a buyer who is served at all has $N(d) \geq 0$; by (MON), $N$ is strictly decreasing in $d$ wherever $\bar{m}$ is strictly increasing, and non-increasing everywhere. The factor $1/[(\sigma-1)d]$ is strictly positive and strictly decreasing in $d$. A non-negative non-increasing function times a positive strictly decreasing function is strictly decreasing, so $\ln \bar{a}$, and hence $\bar{a}$, is strictly decreasing in $d$. □

The cap restricts the optimum directly and holds whatever the shape of the profit function. Serving a buyer further below $\eta$ lowers it; better access raises it (part (i)).

PROOF OF PROPOSITION 6.1. *Step 1 (the cap).* The first sentence of the proposition is Lemma C.5, evaluated at the seller's own access index (C.2); the monotonicity of $\bar{a}$ in the distance of $\tilde{\theta}$ below $\eta$ is Lemma C.5(ii), and its

monotonicity in $T$, hence in $\rho$ for a positive gap, is part (i).

*Step 2 (the fall).* Let the seller's date-$j$ optimum be $(a_j, \theta_j^*)$ at index $T_j$, $j = 0, 1$, and suppose the date-1 optimum serves some $\tilde{\theta} < \eta$. By Lemma C.5 applied at date 1,

$$a_1 \;\le\; \bar{a}(T_1, \tilde{\theta}).$$

If in addition $a_0 > \bar{a}(T_1, \tilde{\theta})$, then $a_1 \le \bar{a}(T_1, \tilde{\theta}) < a_0$, so appeal falls strictly. □

Remark C.6 (Scope of Proposition 6.1). (i) *The condition is sufficient for a fall, not necessary. A seller whose appeal is already at or below $\bar{a}(T_1, \tilde{\theta})$ can serve buyers below $\eta$ without lowering appeal.*

(ii) *The relevant cap is the post-change one. Because $\bar{a}$ rises with access, initial appeal $a_0$ is compared with $\bar{a}(T_1, \tilde{\theta})$, not with $\bar{a}(T_0, \tilde{\theta})$.*

(iii) *The cap bounds appeal only for a specific buyer below $\eta$. Since $\bar{a}(T, \tilde{\theta}) \to \infty$ as $\tilde{\theta} \uparrow \eta$, knowing only that the date-1 cutoff is below $\eta$ gives no bound.*

### C.3. From chosen appeal to the quality index

Lemma C.7 (Quality splits into an appeal term and a revenue term). *Under* (CONF)*, with $\lambda$ the quality index of eq.* (1)*,*

$$\Delta \ln \lambda \;=\; \underbrace{\eta\, \Delta \ln a}_{\text{appeal term}} \;+\; \underbrace{\frac{1}{\sigma - 1}\, \Delta \ln \bar{\Gamma}^*}_{\text{revenue term}}, \qquad \bar{\Gamma}_j^* \equiv \bar{\Gamma}\big(a_j; [\theta_j^*, \bar{\theta}]\big). \tag{C.12}$$

Proof. *Step 1.* Equation (1) is $\ln \lambda = \frac{\sigma}{\sigma-1} \ln p + \frac{1}{\sigma-1} \ln q$, which rearranges to $\ln \lambda = \ln p + \frac{1}{\sigma-1} \ln(pq) = \ln p + \frac{1}{\sigma-1} \ln V$. With one active variety at each date ((CONF)(ii)) the price and revenue in the market are that variety's.

*Step 2.* By (C.1), $\ln p = \eta \ln a + \text{const}$, where the constant depends on $(\sigma, c_0, \varphi)$ and not on $a$ or on the buyers served; by (CONF)(i) it does not move between the dates. By (C.5), $\ln V = \ln \bar{\Gamma}^* + \ln(\sigma c x \tau)$, and the second term does not move between the dates by (CONF)(i) and (iii).

*Step 3.* Substituting Step 2 into Step 1 and differencing, the constants cancel and (C.12) follows. □

By Step 2, the change in $\ln \bar{\Gamma}^*$ equals the change in log revenue, so the second term is the revenue term of Section 6.4.

Lemma C.8 (The revenue term does not fall). *Under* (CONF)*, $\bar{\Gamma}_1^* \ge \bar{\Gamma}_0^*$ for any selection of maximisers at the two dates. Consequently the quality index falls only if chosen appeal falls.*

Proof. *Step 1.* Write $z_j = (a_j, \theta_j^*)$, $\bar{\Gamma}(z)$ for (C.4) at $z$, and $K(z) \equiv F_0 a^\kappa + k\mathsf{N}(\theta^*)$, so that (C.3) reads $\pi(z; T) = T\bar{\Gamma}(z) - K(z)$, affine in $T$ with slope $\bar{\Gamma}(z) \ge 0$. Neither $\bar{\Gamma}$ nor $K$ depends on $T$.

*Step 2.* Optimality of $z_0$ at $T_0$ and of $z_1$ at $T_1$ gives

$$T_0\bar{\Gamma}(z_0) - K(z_0) \;\ge\; T_0\bar{\Gamma}(z_1) - K(z_1), \qquad T_1\bar{\Gamma}(z_1) - K(z_1) \;\ge\; T_1\bar{\Gamma}(z_0) - K(z_0).$$

Adding and cancelling the $K$ terms, $(T_1 - T_0)\left[\bar{\Gamma}(z_1) - \bar{\Gamma}(z_0)\right] \geq 0$, and $T_1 > T_0$ by (CONF)(iv), so $\bar{\Gamma}(z_1) \geq \bar{\Gamma}(z_0)$.

*Step 3.* By Lemma C.7 the second term of (C.12) is then non-negative, so $\Delta \ln \lambda < 0$ requires $\eta \Delta \ln a < 0$, and $\eta > 0$ gives $\Delta \ln a < 0$. □

Corollary C.9 (Revenue does not fall). *Under* (CONF), $\Delta \ln p + \Delta \ln q \geq 0$.

Proof. By (C.5), $\ln V = \ln \bar{\Gamma}^* + \ln(\sigma c x \tau)$, and the second term does not move between the dates by (CONF)(i) and (iii). By Lemma C.8, $\bar{\Gamma}^*_1 \geq \bar{\Gamma}^*_0$. So $\Delta \ln V \geq 0$, and $V = pq$ gives the statement. □

Corollary C.10 (The quality index understates the fall in appeal). *Under* (CONF),

$$\Delta \ln \lambda \;\geq\; \eta\, \Delta \ln a, \qquad \textit{equivalently} \qquad -\Delta \ln \lambda \;\leq\; \eta \left(-\Delta \ln a\right), \tag{C.13}$$

*with equality if and only if $\bar{\Gamma}^*_1 = \bar{\Gamma}^*_0$. In particular, if the quality index falls then $|\Delta \ln \lambda| \leq \eta\, |\Delta \ln a|$.*

Proof. By Lemma C.7, $\Delta \ln \lambda - \eta \Delta \ln a = \frac{1}{\sigma - 1} \Delta \ln \bar{\Gamma}^*$; by Lemma C.8 $\Delta \ln \bar{\Gamma}^* \geq 0$, and $\sigma > 1$, so the right side is non-negative, with equality exactly when $\bar{\Gamma}^*_1 = \bar{\Gamma}^*_0$. If $\Delta \ln \lambda < 0$ then $\eta \Delta \ln a \leq \Delta \ln \lambda < 0$, so both sides of the second form of (C.13) are positive and it is the stated bound on magnitudes. □

Remark C.11 (The revenue implication and the estimates). *Corollary C.9 concerns one seller's revenue in one market; the coefficients of eq. (2) are differences between destination groups. Write $\beta_y$ for the 2002–04 coefficient of eq. (2) for outcome $y$: $y = p$ the log unit value, $y = q$ log quantity and $y = \lambda$ quality.*

An identity. *Log price plus log quantity is log revenue, and the three regressions share their sample, regressors and fixed effects. So $\beta_p + \beta_q$ is the coefficient of a log-revenue regression, and $\beta_\lambda = \beta_p + (\beta_p + \beta_q)/(\sigma - 1)$.*

The baseline specification. *After singletons are removed, every firm × product × year group of the baseline specification holds one US and one non-US observation. Its coefficients are therefore those of a regression of the US-minus-non-US difference on the gap-by-period terms with firm × product and year effects. Write $b_{\mathrm{US}}$ and $b_{\mathrm{NUS}}$ for the coefficients that the residualised regressor would return for log revenue on each leg separately, so that $\beta_p + \beta_q = b_{\mathrm{US}} - b_{\mathrm{NUS}}$. Corollary C.9 signs $b_{\mathrm{US}}$ under three premises.*

(a) (CONF)*(i)–(iii) hold on the US leg for every seller in the sample, and (iv) for every seller with a positive gap. Then $\ln(\sigma c x \tau)$ is fixed (Step 2 of the proof of Lemma C.7), so $b_{\mathrm{US}}$ and $b_{\mathrm{NUS}}$ are the leg slopes themselves.*

(b) *Assumption 2 holds. Then $T_1 = cx$ does not depend on exposure, while $T_0 = cx\, \varpi(\rho_0)$ falls as $\chi$ falls; the gap orders exposure through $(1 + t_2)/(1 + t_1)$ at a given NTR rate $t_1$. Take a seller facing the same index to choose the same maximiser, a selection convention that restricts no primitive. The revealed-preference step in the proof of Lemma C.8 then makes a seller's $\Delta \ln \bar{\Gamma}^*$ non-decreasing in exposure. For sellers with a negative gap, $\chi > 1$ and $T_1 < T_0$, and the same step orders their optima in the other direction.*

(c) *The covariance of the US-leg change in log revenue with the residualised regressor is non-negative. In a balanced panel, (a), (b) and seller primitives distributed independently of the gap imply (c).*

*Under (a)–(c), $b_{\mathrm{US}} \geq 0$, and $\beta_p + \beta_q \geq 0$ if and only if $b_{\mathrm{NUS}} \leq b_{\mathrm{US}}$: exposure must not raise non-US revenue by more than US revenue. The model itself gives $b_{\mathrm{NUS}} = 0$, since PNTR does not change the non-US access index; a non-zero $b_{\mathrm{NUS}}$ is a spillover across legs that the model does not contain.*

*For 2002–04 the baseline sum is $-0.132 + 0.191 = +0.059$ (Table 3, Panels B and C), the sign the premises*

*imply; no standard error is computed for the sum. By the identity the baseline quality coefficient is then no more negative than the price coefficient. It is negative, and so a fall, only for $\sigma$ above about* 1.44.

What a negative baseline sum would mean. *One of the following: exposure raised non-US revenue by more than US revenue;* (CONF)*(i) or (iii) fails on the US leg for some exposed sellers, for instance because entry by other sellers in exposed products lowered the US price index;* (CONF)*(ii) fails for some exposed sellers, which can lower revenue only if the cross-variety terms between their varieties do not vanish; premise (b) or (c) fails; or sampling error.*

The broad specification. *It has no firm × product × year effect and includes firm–products observed on one leg only, so its coefficients do not decompose into two leg slopes, and I derive no sign for its sum. That sum is* $-0.117 + 0.013 = -0.104$ *for 2002–04; if the restriction applied to it, the broad point estimates would violate it. Correspondingly the broad quality coefficient (*$-0.143$*) is larger in absolute value than the price coefficient (*$-0.117$*), since at $\sigma = 5$ the quality coefficient is the price coefficient plus one quarter of the sum. The reading of the model in Section 6 rests on the baseline specification and takes from the broad specification only the sign of the quality coefficient, which is negative for every $\sigma > 1$.*

When a seller reaches down, the two terms of (C.12) move in opposite directions: the appeal term can fall by Proposition 6.1, while the revenue term does not fall. Whether the quality index falls therefore depends on magnitudes. The next proposition gives a sufficient condition.

Proposition C.12 (When the appeal term dominates). *Assume* (MON), (NORM), (BND) *and* (CONF). *Suppose the date-1 optimum serves a buyer $\tilde{\theta} < \eta$, write $\bar{a} \equiv \bar{a}(T_1, \tilde{\theta})$ for the cap of* (C.10) *at date 1, and let $\bar{\Gamma}_\Theta(a) \equiv \bar{\Gamma}(a; \Theta)$ be the value of serving every buyer. Define*

$$\mathcal{U} \;\equiv\; \eta \left[\ln \bar{a} - \ln a_0\right] \;+\; \frac{1}{\sigma - 1} \ln \frac{\max_{1 \le a \le \bar{a}} \bar{\Gamma}_\Theta(a)}{\bar{\Gamma}_0^*}. \tag{C.14}$$

*Then $\Delta \ln \lambda \le \mathcal{U}$. In particular, if $\mathcal{U} < 0$ then the quality index falls.*

Proof. *Step 1 (the appeal term).* Lemma C.5 at date 1 gives $a_1 \le \bar{a}$. Since $\eta > 0$, $\eta \Delta \ln a = \eta[\ln a_1 - \ln a_0] \le \eta[\ln \bar{a} - \ln a_0]$.

*Step 2 (the revenue term).* The integrand of (C.4) is positive, so enlarging the set of buyers enlarges the integral: $\bar{\Gamma}_1^* = \bar{\Gamma}(a_1; [\theta_1^*, \bar{\theta}]) \le \bar{\Gamma}_\Theta(a_1)$. By (NORM) and Step 1, $a_1 \in [1, \bar{a}]$, so $\bar{\Gamma}_\Theta(a_1) \le \max_{1 \le a \le \bar{a}} \bar{\Gamma}_\Theta(a)$. The maximum is attained because $\bar{\Gamma}_\Theta$ is continuous on the compact interval $[1, \bar{a}]$.

*Step 3.* Substituting Steps 1 and 2 into (C.12) gives $\Delta \ln \lambda \le \mathcal{U}$. If $\mathcal{U} < 0$ then $\Delta \ln \lambda < 0$. □

Remark C.13 (The maximum in (C.14)). *The second term of* (C.14) *is a maximum over $[1, \bar{a}]$, not the value at $a = \bar{a}$. By* (C.6) *on $S = \Theta$,*

$$\frac{d\bar{\Gamma}_\Theta(a)}{d \ln a} = (\sigma - 1) \int_\Theta (\theta - \eta)\, \bar{c}\, \bar{m}(\theta)\, a^{(\sigma-1)(\theta-\eta)} g(\theta)\, d\theta,$$

*whose sign is that of the spending-weighted average of $(\theta - \eta)$ over all types. If enough spending lies below $\eta$, $\bar{\Gamma}_\Theta$ is decreasing on $[1, \bar{a}]$, its maximum is $\bar{\Gamma}_\Theta(1) = \int_\Theta \bar{c}\, \bar{m}\, g$, and evaluating at $\bar{a}$ would give the minimum and no bound.*

REMARK C.14 (Scope of Proposition C.12). (i) *The condition is sufficient, not necessary: the bound replaces the date-1 appeal and the date-1 set of buyers by upper bounds, so a seller with* $\mathcal{U} > 0$ *can still show a fall.*

(ii) *It requires a specific buyer below* $\eta$ *known to be served at date 1, for the reason in Remark C.6(iii).*

(iii) *It concerns one seller under* (CONF). *It does not cover a seller that opens a second variety, and it does not aggregate across sellers without an argument about their distribution.*

## Appendix D. Data construction

This appendix states the definitions behind Section 2. Counts of the estimation samples are in Table 1.

### D.1. From customs records to firm–products

The source is the transaction-level export records of the Chinese customs administration for 2000 to 2014. Imports are dropped first. A record with a tariff code that does not resolve to a six-digit product is removed. Records are summed to a firm, an HS6 product, a destination group and a year. A destination string that the country concordance does not recognize is assigned to the non-US group and not discarded, so the non-US group is the complement of the US group and not a list. The US group is the United States and its territories as coded in the country concordance.

I use *firm–product* for a firm's sales of one HS6 product, and *firm–product–destination* when the destination group matters. The destination group has two values, so the non-US side of a firm–product pools every other destination the firm serves in that product and year. The destination × year effects therefore absorb everything common to US shipments in a year and everything common to non-US shipments in a year, and nothing finer. A shock to one comparison destination is absorbed only through its weight in the non-US average, which is why Section 5 checks the comparison with destinations kept separate.

### D.2. Price, quantity and the unit choice

The records report a value and a value per physical unit. The quantity of a record is their ratio wherever the value per unit is strictly positive. The price of a firm–product–destination in a year is total value over total quantity, summed across lines. A record with no usable quantity enters neither the numerator nor the denominator, since leaving it out of the quantity sum alone would bias the price upward. Quantities can be added only if the lines share a physical unit, so within each product and year I keep the records in the unit code that carries the most export value in that product and year, and drop the others. A firm–product–destination with no usable quantity, or with a non-positive quantity, is dropped and not carried with a missing outcome. I winsorize log price and log quantity at their 1st and 99th percentiles within each HS6 product and year, pooling the two destination groups and replacing values without dropping rows. Quality is formed from the winsorized values by eq. (1).

### D.3. The NTR gap and the change of tariff revision

The gap of a product is its column-2 ad valorem rate minus its most-favored-nation (NTR) ad valorem rate in 2000, entered as a proportion; Handley and Limão (2017) also use the rates of 2000. Of the 3,794 products of the estimation sample, 304 have a gap of exactly zero and 5 a slightly negative one. I take the rates from the replication package of Handley and Limão (2017), which carries them at the six-digit level of the 1996 revision of the Harmonized System (HS). The customs lines arrive in the HS revision in force in each year and are mapped to six-digit codes of the 1988/92 revision, the level at which I keep products. The published mapping from the 1996 to the 1988/92 revision is many-to-one: no 1996 code maps to two 1988/92 codes, and 116 of the 1988/92 codes have more than one 1996 code. The gap of a 1988/92 code is the unweighted average of the gaps of the 1996 codes that map to it. Weighting those codes by pre-period US imports instead changes the measure only for

the codes with several 1996 codes. Products with no gap are not in the sample. Section 2 gives the distribution of the gap.

### D.4. Continuing relationships and 2006

Relationships and the continuing sample are defined in Section 2, and the baseline sample in Section 3; the base period is 2000–01 and the post period starts in 2002. I drop 2006 for price, quantity and quality, for the reason given in Section 2. For the same firm, HS8 line and country, the median log unit value rises by 1.19 from 2005 to 2006, a factor of about 3.3, and falls by 0.85 from 2006 to 2007, while log value moves by +0.09 and +0.13. BACI China-to-US log unit values change by −0.002 over 2005–06, where the customs log unit values change by +1.69 for the US group and +1.34 for the non-US group. Retaining 2006 leaves the 2002–04 baseline estimate at −0.119 (0.039) but moves the 2005–14 estimate from −0.230 (0.073) to −0.094 (0.080), which is no longer significant. The 2005–14 estimate is therefore sensitive to the treatment of 2006.